**Newman-Penrose Formalism Calculations for**

**metric:**

*Name:* Stephani et al., Equation (13.57)

*Reference(s):* [1]

*Line element:*

$$ds^2 = -2\,\epsilon k^2 W\left(x_0\right)^n \cosh\left(x_0\right)\sinh\left(x_0\right)\mathrm{d}x_0 \otimes \mathrm{d}x_0 + 2\,\epsilon k^2 W\left(x_0\right)^n \cosh\left(x_0\right)\sinh\left(x_0\right)\mathrm{d}x_1 \otimes \mathrm{d}x_1 + 2\,k^2 W\left(x_0\right)\cosh\left(x_0\right) e^{\left(2\,n x_1 + 2\,x_1\right)}\sinh\left(x_0\right)\mathrm{d}x_2 \otimes \mathrm{d}x_2 + \frac{2\,k^2\cosh\left(x_0\right) e^{\left(-2\,n x_1 + 2\,x_1\right)}\sinh\left(x_0\right)}{W\left(x_0\right)}\mathrm{d}x_3 \otimes \mathrm{d}x_3 \tag{1}$$

*Coordinates:* $(x_0, x_1, x_2, x_3)$

**Null tetrad components:**

*Covariant components:*

$$l = \left(\frac{1}{2}\sqrt{2}\sqrt{\epsilon W\left(x_0\right)^n \sinh\left(2\,x_0\right)}k\right) dx_0 + \left(\frac{1}{2}\sqrt{2}\sqrt{\epsilon W\left(x_0\right)^n \sinh\left(2\,x_0\right)}k\right) dx_1 + (0)\,dx_2 + (0)\,dx_3 \tag{2}$$

$$n = \left(\frac{1}{2}\sqrt{2}\sqrt{\epsilon W\left(x_0\right)^n \sinh\left(2\,x_0\right)}k\right) dx_0 + \left(-\frac{1}{2}\sqrt{2}\sqrt{\epsilon W\left(x_0\right)^n \sinh\left(2\,x_0\right)}k\right) dx_1 + (0)\,dx_2 + (0)\,dx_3 \tag{3}$$

$$m = (0)\,dx_0 + (0)\,dx_1 + \left(\frac{1}{2}\sqrt{2}k\sqrt{W\left(x_0\right)}e^{\left((n+1)x_1\right)}\sqrt{\sinh\left(2\,x_0\right)}\right) dx_2 + \left(\frac{i\sqrt{2}ke^{\left(-(n-1)x_1\right)}\sqrt{\sinh\left(2\,x_0\right)}}{2\sqrt{W\left(x_0\right)}}\right) dx_3 \tag{4}$$

$$\overline{m} = (0)\,dx_0 + (0)\,dx_1 + \left(\frac{1}{2}\sqrt{2}k\sqrt{W\left(x_0\right)}e^{\left((n+1)x_1\right)}\sqrt{\sinh\left(2\,x_0\right)}\right) dx_2 + \left(-\frac{i\sqrt{2}ke^{\left(-(n-1)x_1\right)}\sqrt{\sinh\left(2\,x_0\right)}}{2\sqrt{W\left(x_0\right)}}\right) dx_3 \tag{5}$$

*Contravariant components:*

$$l = \left(-\frac{1}{2\sqrt{\epsilon}k\sqrt{W\left(x_0\right)^n}\sqrt{\cosh\left(x_0\right)}\sqrt{\sinh\left(x_0\right)}}\right)dx_0 + \left(\frac{1}{2\sqrt{\epsilon}k\sqrt{W\left(x_0\right)^n}\sqrt{\cosh\left(x_0\right)}\sqrt{\sinh\left(x_0\right)}}\right)dx_1 + (0)\,dx_2 + (0)\,dx_3 \tag{6}$$

$$n = \left(-\frac{1}{2\sqrt{\epsilon}k\sqrt{W\left(x_0\right)^n}\sqrt{\cosh\left(x_0\right)}\sqrt{\sinh\left(x_0\right)}}\right)dx_0 + \left(-\frac{1}{2\sqrt{\epsilon}k\sqrt{W\left(x_0\right)^n}\sqrt{\cosh\left(x_0\right)}\sqrt{\sinh\left(x_0\right)}}\right)dx_1 + (0)\,dx_2 + (0)\,dx_3 \tag{7}$$

$$m = (0)\,dx_0 + (0)\,dx_1 + \left(\frac{e^{(-nx_1-x_1)}}{2\,k\sqrt{W\left(x_0\right)}\sqrt{\cosh\left(x_0\right)}\sqrt{\sinh\left(x_0\right)}}\right)dx_2 + \left(\frac{i\sqrt{W\left(x_0\right)}e^{(nx_1-x_1)}}{2\,k\sqrt{\cosh\left(x_0\right)}\sqrt{\sinh\left(x_0\right)}}\right)dx_3 \tag{8}$$

$$\overline{m} = (0)\,dx_0 + (0)\,dx_1 + \left(\frac{e^{(-nx_1-x_1)}}{2\,k\sqrt{W\left(x_0\right)}\sqrt{\cosh\left(x_0\right)}\sqrt{\sinh\left(x_0\right)}}\right)dx_2 + \left(-\frac{i\sqrt{W\left(x_0\right)}e^{(nx_1-x_1)}}{2\,k\sqrt{\cosh\left(x_0\right)}\sqrt{\sinh\left(x_0\right)}}\right)dx_3 \tag{9}$$

**Directional derivatives:**

$$D = -\frac{\frac{\partial}{\partial x_0} f\left(x_0, x_1, x_2, x_3\right)}{2\sqrt{\epsilon} k \sqrt{W\left(x_0\right)^n} \sqrt{\cosh\left(x_0\right)} \sqrt{\sinh\left(x_0\right)}} + \frac{\frac{\partial}{\partial x_1} f\left(x_0, x_1, x_2, x_3\right)}{2\sqrt{\epsilon} k \sqrt{W\left(x_0\right)^n} \sqrt{\cosh\left(x_0\right)} \sqrt{\sinh\left(x_0\right)}} \tag{10}$$

$$\Delta = -\frac{\frac{\partial}{\partial x_0} f\left(x_0, x_1, x_2, x_3\right)}{2\sqrt{\epsilon} k \sqrt{W\left(x_0\right)^n} \sqrt{\cosh\left(x_0\right)} \sqrt{\sinh\left(x_0\right)}} - \frac{\frac{\partial}{\partial x_1} f\left(x_0, x_1, x_2, x_3\right)}{2\sqrt{\epsilon} k \sqrt{W\left(x_0\right)^n} \sqrt{\cosh\left(x_0\right)} \sqrt{\sinh\left(x_0\right)}} \tag{11}$$

$$\delta = \frac{i\sqrt{W\left(x_0\right)} e^{\left(n x_1 - x_1\right)} \frac{\partial}{\partial x_3} f\left(x_0, x_1, x_2, x_3\right)}{2\, k \sqrt{\cosh\left(x_0\right)} \sqrt{\sinh\left(x_0\right)}} + \frac{e^{\left(-n x_1 - x_1\right)} \frac{\partial}{\partial x_2} f\left(x_0, x_1, x_2, x_3\right)}{2\, k \sqrt{W\left(x_0\right)} \sqrt{\cosh\left(x_0\right)} \sqrt{\sinh\left(x_0\right)}} \tag{12}$$

$$\overline{\delta} = -\frac{i\sqrt{W\left(x_0\right)} e^{\left(n x_1 - x_1\right)} \frac{\partial}{\partial x_3} f\left(x_0, x_1, x_2, x_3\right)}{2\, k \sqrt{\cosh\left(x_0\right)} \sqrt{\sinh\left(x_0\right)}} + \frac{e^{\left(-n x_1 - x_1\right)} \frac{\partial}{\partial x_2} f\left(x_0, x_1, x_2, x_3\right)}{2\, k \sqrt{W\left(x_0\right)} \sqrt{\cosh\left(x_0\right)} \sqrt{\sinh\left(x_0\right)}} \tag{13}$$

**Spin coefficients:**

$$\rho = \frac{\sqrt{\cosh(x_0)}}{2\sqrt{\epsilon}k\sqrt{W(x_0)^n}\sinh(x_0)^{\frac{3}{2}}} - \frac{1}{2\sqrt{\epsilon}k\sqrt{W(x_0)^n}\sqrt{\cosh(x_0)}\sqrt{\sinh(x_0)}} - \frac{1}{4\sqrt{\epsilon}k\sqrt{W(x_0)^n}\cosh(x_0)^{\frac{3}{2}}\sinh(x_0)^{\frac{3}{2}}} \tag{14}$$

$$\sigma = -\frac{n}{2\sqrt{\epsilon}k\sqrt{W(x_0)^n}\sqrt{\cosh(x_0)}\sqrt{\sinh(x_0)}} + \frac{\frac{\partial}{\partial x_0}W(x_0)}{4\sqrt{\epsilon}k\sqrt{W(x_0)^n}W(x_0)\sqrt{\cosh(x_0)}\sqrt{\sinh(x_0)}} \tag{15}$$

$$\kappa = 0 \tag{16}$$

$$\mu = -\frac{\sqrt{\cosh(x_0)}}{2\sqrt{\epsilon}k\sqrt{W(x_0)^n}\sinh(x_0)^{\frac{3}{2}}} - \frac{1}{2\sqrt{\epsilon}k\sqrt{W(x_0)^n}\sqrt{\cosh(x_0)}\sqrt{\sinh(x_0)}} + \frac{1}{4\sqrt{\epsilon}k\sqrt{W(x_0)^n}\cosh(x_0)^{\frac{3}{2}}\sinh(x_0)^{\frac{3}{2}}} \tag{17}$$

$$\lambda = -\frac{2nW(x_0) + \frac{\partial}{\partial x_0}W(x_0)}{4\sqrt{\epsilon}k\sqrt{W(x_0)^n}W(x_0)\sqrt{\cosh(x_0)}\sqrt{\sinh(x_0)}} \tag{18}$$

$$\nu = 0 \tag{19}$$

$$\alpha = 0 \tag{20}$$

$$\beta = 0 \tag{21}$$

$$\tau = 0 \tag{22}$$

$$\pi = 0 \tag{23}$$

$$\epsilon = -\frac{n\frac{\partial}{\partial x_0}W\left(x_0\right)}{8\sqrt{\epsilon}k\sqrt{W\left(x_0\right)^n}W\left(x_0\right)\sqrt{\cosh\left(x_0\right)}\sqrt{\sinh\left(x_0\right)}} - \frac{\sqrt{\sinh\left(x_0\right)}}{4\sqrt{\epsilon}k\sqrt{W\left(x_0\right)^n}\cosh\left(x_0\right)^{\frac{3}{2}}} - \frac{1}{8\sqrt{\epsilon}k\sqrt{W\left(x_0\right)^n}\cosh\left(x_0\right)^{\frac{3}{2}}\sinh\left(x_0\right)^{\frac{3}{2}}} \tag{24}$$

$$\gamma = \frac{n\frac{\partial}{\partial x_0}W\left(x_0\right)}{8\sqrt{\epsilon}k\sqrt{W\left(x_0\right)^n}W\left(x_0\right)\sqrt{\cosh\left(x_0\right)}\sqrt{\sinh\left(x_0\right)}} + \frac{\sqrt{\sinh\left(x_0\right)}}{4\sqrt{\epsilon}k\sqrt{W\left(x_0\right)^n}\cosh\left(x_0\right)^{\frac{3}{2}}} + \frac{1}{8\sqrt{\epsilon}k\sqrt{W\left(x_0\right)^n}\cosh\left(x_0\right)^{\frac{3}{2}}\sinh\left(x_0\right)^{\frac{3}{2}}} \tag{25}$$

**Relations between spin coefficients:**

**Ricci tensor components:**

$$\Phi_{00} = -\frac{n^2 W(x_0)^{-n}\cosh(x_0)}{4\,\epsilon k^2 \sinh(x_0)^3} + \frac{nW(x_0)^{-n}\frac{\partial}{\partial x_0}W(x_0)}{4\,\epsilon k^2 W(x_0)\sinh(x_0)^2} + \frac{n^2 W(x_0)^{-n}}{4\,\epsilon k^2\cosh(x_0)\sinh(x_0)^3} - \frac{W(x_0)^{-n}\cosh(x_0)\frac{\partial}{\partial x_0}W(x_0)^2}{16\,\epsilon k^2 W(x_0)^2\sinh(x_0)^3} - \frac{nW(x_0)^{-n}\frac{\partial}{\partial x_0}W(x_0)}{8\,\epsilon k^2 W(x_0)\cosh(x_0)^2\sinh(x_0)^2} + \frac{W(x_0)^{-n}\frac{\partial}{\partial x_0}W(x_0)^2}{16\,\epsilon k^2 W(x_0)^2\cosh(x_0)\sinh(x_0)^3} + \frac{3\,W(x_0)^{-n}}{16\,\epsilon k^2\cosh(x_0)^3\sinh(x_0)^3} \tag{26}$$

$$\Phi_{01} = 0 \tag{27}$$

$$\Phi_{02} = -\frac{nW(x_0)^{-n}}{2\,\epsilon k^2\cosh(x_0)\sinh(x_0)} + \frac{W(x_0)^{-n}\frac{\partial}{\partial x_0}W(x_0)}{4\,\epsilon k^2 W(x_0)\cosh(x_0)^2} - \frac{W(x_0)^{-n}\frac{\partial}{\partial x_0}W(x_0)^2}{8\,\epsilon k^2 W(x_0)^2\cosh(x_0)\sinh(x_0)} + \frac{W(x_0)^{-n}\frac{\partial^2}{(\partial x_0)^2}W(x_0)}{8\,\epsilon k^2 W(x_0)\cosh(x_0)\sinh(x_0)} + \frac{W(x_0)^{-n}\frac{\partial}{\partial x_0}W(x_0)}{8\,\epsilon k^2 W(x_0)\cosh(x_0)^2\sinh(x_0)^2} \tag{28}$$

$$\Phi_{10} = 0 \tag{29}$$

$$\Phi_{11} = \frac{n^2 W(x_0)^{-n}\cosh(x_0)}{8\,\epsilon k^2\sinh(x_0)^3} + \frac{nW(x_0)^{-n}\cosh(x_0)\frac{\partial}{\partial x_0}W(x_0)^2}{16\,\epsilon k^2 W(x_0)^2\sinh(x_0)^3} - \frac{nW(x_0)^{-n}\cosh(x_0)\frac{\partial^2}{(\partial x_0)^2}W(x_0)}{16\,\epsilon k^2 W(x_0)\sinh(x_0)^3} - \frac{n^2 W(x_0)^{-n}}{8\,\epsilon k^2\cosh(x_0)\sinh(x_0)^3} - \frac{W(x_0)^{-n}\cosh(x_0)\frac{\partial}{\partial x_0}W(x_0)^2}{32\,\epsilon k^2 W(x_0)^2\sinh(x_0)^3} - \frac{nW(x_0)^{-n}\frac{\partial}{\partial x_0}W(x_0)^2}{16\,\epsilon k^2 W(x_0)^2\cosh(x_0)\sinh(x_0)^3} + \frac{nW(x_0)^{-n}\frac{\partial^2}{(\partial x_0)^2}W(x_0)}{16\,\epsilon k^2 W(x_0)\cosh(x_0)\sinh(x_0)^3} + \frac{W(x_0)^{-n}\frac{\partial}{\partial x_0}W(x_0)^2}{32\,\epsilon k^2 W(x_0)^2\cosh(x_0)\sinh(x_0)^3} + \frac{3\,W(x_0)^{-n}}{32\,\epsilon k^2\cosh(x_0)^3\sinh(x_0)^3} \tag{30}$$

$$\Phi_{12} = 0 \tag{31}$$

$$\Phi_{20} = -\frac{nW(x_0)^{-n}}{2\,\epsilon k^2 \cosh(x_0)\sinh(x_0)} + \frac{W(x_0)^{-n}\,\frac{\partial}{\partial x_0}W(x_0)}{4\,\epsilon k^2 W(x_0)\cosh(x_0)^2} - \frac{W(x_0)^{-n}\,\frac{\partial}{\partial x_0}W(x_0)^2}{8\,\epsilon k^2 W(x_0)^2\cosh(x_0)\sinh(x_0)} + \frac{W(x_0)^{-n}\,\frac{\partial^2}{(\partial x_0)^2}W(x_0)}{8\,\epsilon k^2 W(x_0)\cosh(x_0)\sinh(x_0)} + \frac{W(x_0)^{-n}\,\frac{\partial}{\partial x_0}W(x_0)}{8\,\epsilon k^2 W(x_0)\cosh(x_0)^2\sinh(x_0)^2} \tag{32}$$

$$\Phi_{21} = 0 \tag{33}$$

$$\begin{aligned}\Phi_{22} = &-\frac{n^2 W(x_0)^{-n}\cosh(x_0)}{4\,\epsilon k^2\sinh(x_0)^3} + \frac{nW(x_0)^{-n}\,\frac{\partial}{\partial x_0}W(x_0)}{4\,\epsilon k^2 W(x_0)\sinh(x_0)^2} + \frac{n^2 W(x_0)^{-n}}{4\,\epsilon k^2\cosh(x_0)\sinh(x_0)^3} - \frac{W(x_0)^{-n}\cosh(x_0)\,\frac{\partial}{\partial x_0}W(x_0)^2}{16\,\epsilon k^2 W(x_0)^2\sinh(x_0)^3} \\ &- \frac{nW(x_0)^{-n}\,\frac{\partial}{\partial x_0}W(x_0)}{8\,\epsilon k^2 W(x_0)\cosh(x_0)^2\sinh(x_0)^2} + \frac{W(x_0)^{-n}\,\frac{\partial}{\partial x_0}W(x_0)^2}{16\,\epsilon k^2 W(x_0)^2\cosh(x_0)\sinh(x_0)^3} + \frac{3\,W(x_0)^{-n}}{16\,\epsilon k^2\cosh(x_0)^3\sinh(x_0)^3}\end{aligned} \tag{34}$$

$$\begin{aligned}\Lambda = &-\frac{n^2 W(x_0)^{-n}\cosh(x_0)}{24\,\epsilon k^2\sinh(x_0)^3} - \frac{nW(x_0)^{-n}\cosh(x_0)\,\frac{\partial}{\partial x_0}W(x_0)^2}{48\,\epsilon k^2 W(x_0)^2\sinh(x_0)^3} + \frac{nW(x_0)^{-n}\cosh(x_0)\,\frac{\partial^2}{(\partial x_0)^2}W(x_0)}{48\,\epsilon k^2 W(x_0)\sinh(x_0)^3} \\ &+ \frac{n^2 W(x_0)^{-n}}{24\,\epsilon k^2\cosh(x_0)\sinh(x_0)^3} + \frac{W(x_0)^{-n}\cosh(x_0)\,\frac{\partial}{\partial x_0}W(x_0)^2}{96\,\epsilon k^2 W(x_0)^2\sinh(x_0)^3} + \frac{nW(x_0)^{-n}\,\frac{\partial}{\partial x_0}W(x_0)^2}{48\,\epsilon k^2 W(x_0)^2\cosh(x_0)\sinh(x_0)^3} \\ &- \frac{nW(x_0)^{-n}\,\frac{\partial^2}{(\partial x_0)^2}W(x_0)}{48\,\epsilon k^2 W(x_0)\cosh(x_0)\sinh(x_0)^3} - \frac{W(x_0)^{-n}\,\frac{\partial}{\partial x_0}W(x_0)^2}{96\,\epsilon k^2 W(x_0)^2\cosh(x_0)\sinh(x_0)^3} - \frac{W(x_0)^{-n}}{32\,\epsilon k^2\cosh(x_0)^3\sinh(x_0)^3}\end{aligned} \tag{35}$$

**Weyl tensor components:**

$$\Psi_0 = -\frac{n^2 W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)}{4\,\epsilon k^2 W(x_0)\cosh(x_0)\sinh(x_0)} - \frac{n W(x_0)^{-n}}{2\,\epsilon k^2 \cosh(x_0)\sinh(x_0)} + \frac{n W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)^2}{8\,\epsilon k^2 W(x_0)^2 \cosh(x_0)\sinh(x_0)} + \frac{W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)}{4\,\epsilon k^2 W(x_0)\cosh(x_0)\sinh(x_0)} + \frac{W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)^2}{8\,\epsilon k^2 W(x_0)^2 \cosh(x_0)\sinh(x_0)} - \frac{W(x_0)^{-n} \frac{\partial^2}{(\partial x_0)^2} W(x_0)}{8\,\epsilon k^2 W(x_0)\cosh(x_0)\sinh(x_0)} \tag{36}$$

$$\Psi_1 = 0 \tag{37}$$

$$\Psi_2 = -\frac{n^2 W(x_0)^{-n}}{6\,\epsilon k^2 \cosh(x_0)\sinh(x_0)} + \frac{n W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)^2}{24\,\epsilon k^2 W(x_0)^2 \cosh(x_0)\sinh(x_0)} - \frac{n W(x_0)^{-n} \frac{\partial^2}{(\partial x_0)^2} W(x_0)}{24\,\epsilon k^2 W(x_0)\cosh(x_0)\sinh(x_0)} + \frac{W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)^2}{24\,\epsilon k^2 W(x_0)^2 \cosh(x_0)\sinh(x_0)} \tag{38}$$

$$\Psi_3 = 0 \tag{39}$$

$$\Psi_4 = \frac{n^2 W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)}{4\,\epsilon k^2 W(x_0)\cosh(x_0)\sinh(x_0)} - \frac{n W(x_0)^{-n}}{2\,\epsilon k^2 \cosh(x_0)\sinh(x_0)} + \frac{n W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)^2}{8\,\epsilon k^2 W(x_0)^2 \cosh(x_0)\sinh(x_0)} - \frac{W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)}{4\,\epsilon k^2 W(x_0)\cosh(x_0)\sinh(x_0)} + \frac{W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)^2}{8\,\epsilon k^2 W(x_0)^2 \cosh(x_0)\sinh(x_0)} - \frac{W(x_0)^{-n} \frac{\partial^2}{(\partial x_0)^2} W(x_0)}{8\,\epsilon k^2 W(x_0)\cosh(x_0)\sinh(x_0)} \tag{40}$$

**Newman-Penrose equations:**

$$NP1 = 0 \tag{41}$$

$$NP2 = 0 \tag{42}$$

$$NP3 = 0 \tag{43}$$

$$NP4 = 0 \tag{44}$$

$$NP5 = 0 \tag{45}$$

$$NP6 = 0 \tag{46}$$

$$NP7 = 0 \tag{47}$$

$$NP8 = 0 \tag{48}$$

$$NP9 = 0 \tag{49}$$

$$NP10 = 0 \tag{50}$$

$$NP11 = 0 \tag{51}$$

$$NP12 = 0 \tag{52}$$

$$NP13 = 0 \tag{53}$$

$$NP14 = 0 \tag{54}$$

$$NP15 = 0 \tag{55}$$

$$NP16 = 0 \tag{56}$$

$$NP17 = 0 \tag{57}$$

$$NP18 = 0 \tag{58}$$

**Bianchi identities:**

$$BI1 = 0 \tag{59}$$

$$BI2 = 0 \tag{60}$$

$$BI3 = 0 \tag{61}$$

$$BI4 = 0 \tag{62}$$

$$BI5 = 0 \tag{63}$$

$$BI6 = 0 \tag{64}$$

$$BI7 = 0 \tag{65}$$

$$BI8 = 0 \tag{66}$$

$$BI9 = 0 \tag{67}$$

$$BI10 = 0 \tag{68}$$

$$BI11 = 0 \tag{69}$$

**Petrov classification:**

*Petrov class:*

A Type transformations:

*Spin coefficients:*

$$\rho' = \rho \tag{70}$$

$$\sigma' = \sigma \tag{71}$$

$$\kappa' = 0 \tag{72}$$

$$\begin{aligned}\mu' = &-\frac{nz^2}{2\sqrt{\epsilon}k\sqrt{W(x_0)^n}\sqrt{\cosh(x_0)}\sqrt{\sinh(x_0)}} - \frac{nz^2\frac{\partial}{\partial x_0}W(x_0)}{4\sqrt{\epsilon}k\sqrt{W(x_0)^n}W(x_0)\sqrt{\cosh(x_0)}\sqrt{\sinh(x_0)}} \\ &- \frac{z^2\sqrt{\sinh(x_0)}}{2\sqrt{\epsilon}k\sqrt{W(x_0)^n}\cosh(x_0)^{\frac{3}{2}}} + \frac{z^2\frac{\partial}{\partial x_0}W(x_0)}{4\sqrt{\epsilon}k\sqrt{W(x_0)^n}W(x_0)\sqrt{\cosh(x_0)}\sqrt{\sinh(x_0)}} + \frac{z\sqrt{\cosh(x_0)}}{\sqrt{\epsilon}k\sqrt{W(x_0)^n}\sinh(x_0)^{\frac{3}{2}}} \\ &- \frac{z}{\sqrt{\epsilon}k\sqrt{W(x_0)^n}\sqrt{\cosh(x_0)}\sqrt{\sinh(x_0)}} - \frac{z^2}{4\sqrt{\epsilon}k\sqrt{W(x_0)^n}\cosh(x_0)^{\frac{3}{2}}\sinh(x_0)^{\frac{3}{2}}} - \frac{\sqrt{\cosh(x_0)}}{2\sqrt{\epsilon}k\sqrt{W(x_0)^n}\sinh(x_0)^{\frac{3}{2}}} \\ &- \frac{1}{2\sqrt{\epsilon}k\sqrt{W(x_0)^n}\sqrt{\cosh(x_0)}\sqrt{\sinh(x_0)}} - \frac{z}{2\sqrt{\epsilon}k\sqrt{W(x_0)^n}\cosh(x_0)^{\frac{3}{2}}\sinh(x_0)^{\frac{3}{2}}} + \frac{1}{4\sqrt{\epsilon}k\sqrt{W(x_0)^n}\cosh(x_0)^{\frac{3}{2}}\sinh(x_0)^{\frac{3}{2}}}\end{aligned} \tag{73}$$

$$\lambda' = \lambda \tag{74}$$

$$\begin{aligned}\nu' = &-\frac{nz^2\frac{\partial}{\partial x_0}W(x_0)}{4\sqrt{\epsilon}k\sqrt{W(x_0)^n}W(x_0)\sqrt{\cosh(x_0)}\sqrt{\sinh(x_0)}} + \frac{z^2\sqrt{\cosh(x_0)}}{2\sqrt{\epsilon}k\sqrt{W(x_0)^n}\sinh(x_0)^{\frac{3}{2}}} \\ &- \frac{z^2}{2\sqrt{\epsilon}k\sqrt{W(x_0)^n}\sqrt{\cosh(x_0)}\sqrt{\sinh(x_0)}} - \frac{z^2\sqrt{\sinh(x_0)}}{2\sqrt{\epsilon}k\sqrt{W(x_0)^n}\cosh(x_0)^{\frac{3}{2}}} - \frac{z^2}{2\sqrt{\epsilon}k\sqrt{W(x_0)^n}\cosh(x_0)^{\frac{3}{2}}\sinh(x_0)^{\frac{3}{2}}}\end{aligned} \tag{75}$$

$$\begin{aligned}\alpha' = &-\frac{nz\frac{\partial}{\partial x_0}W(x_0)}{8\sqrt{\epsilon}k\sqrt{W(x_0)^n}W(x_0)\sqrt{\cosh(x_0)}\sqrt{\sinh(x_0)}} + \frac{z\sqrt{\cosh(x_0)}}{2\sqrt{\epsilon}k\sqrt{W(x_0)^n}\sinh(x_0)^{\frac{3}{2}}} \\ &- \frac{z}{2\sqrt{\epsilon}k\sqrt{W(x_0)^n}\sqrt{\cosh(x_0)}\sqrt{\sinh(x_0)}} - \frac{z\sqrt{\sinh(x_0)}}{4\sqrt{\epsilon}k\sqrt{W(x_0)^n}\cosh(x_0)^{\frac{3}{2}}} - \frac{3z}{8\sqrt{\epsilon}k\sqrt{W(x_0)^n}\cosh(x_0)^{\frac{3}{2}}\sinh(x_0)^{\frac{3}{2}}}\end{aligned} \tag{76}$$

$$\beta' = -\frac{nz}{2\sqrt{\epsilon}k\sqrt{W(x_0)^n}\sqrt{\cosh(x_0)}\sqrt{\sinh(x_0)}} - \frac{n\overline{z}\frac{\partial}{\partial x_0}W(x_0)}{8\sqrt{\epsilon}k\sqrt{W(x_0)^n}W(x_0)\sqrt{\cosh(x_0)}\sqrt{\sinh(x_0)}} - \frac{\overline{z}\sqrt{\sinh(x_0)}}{4\sqrt{\epsilon}k\sqrt{W(x_0)^n}\cosh(x_0)^{\frac{3}{2}}} + \frac{z\frac{\partial}{\partial x_0}W(x_0)}{4\sqrt{\epsilon}k\sqrt{W(x_0)^n}W(x_0)\sqrt{\cosh(x_0)}\sqrt{\sinh(x_0)}} - \frac{\overline{z}}{8\sqrt{\epsilon}k\sqrt{W(x_0)^n}\cosh(x_0)^{\frac{3}{2}}\sinh(x_0)^{\frac{3}{2}}} \tag{77}$$

$$\tau' = -\frac{nz\frac{\partial}{\partial x_0}W(x_0)}{8\sqrt{\epsilon}k\sqrt{W(x_0)^n}W(x_0)\sqrt{\cosh(x_0)}\sqrt{\sinh(x_0)}} - \frac{z\sqrt{\sinh(x_0)}}{4\sqrt{\epsilon}k\sqrt{W(x_0)^n}\cosh(x_0)^{\frac{3}{2}}} - \frac{z}{8\sqrt{\epsilon}k\sqrt{W(x_0)^n}\cosh(x_0)^{\frac{3}{2}}\sinh(x_0)^{\frac{3}{2}}} \tag{78}$$

$$\pi' = -\frac{nz}{2\sqrt{\epsilon}k\sqrt{W(x_0)^n}\sqrt{\cosh(x_0)}\sqrt{\sinh(x_0)}} + \frac{z\frac{\partial}{\partial x_0}W(x_0)}{4\sqrt{\epsilon}k\sqrt{W(x_0)^n}W(x_0)\sqrt{\cosh(x_0)}\sqrt{\sinh(x_0)}} \tag{79}$$

$$\epsilon' = \epsilon \tag{80}$$

$$\begin{aligned}\gamma' = &-\frac{nz^2}{2\sqrt{\epsilon}k\sqrt{W(x_0)^n}\sqrt{\cosh(x_0)}\sqrt{\sinh(x_0)}} - \frac{nz\overline{z}\frac{\partial}{\partial x_0}W(x_0)}{8\sqrt{\epsilon}k\sqrt{W(x_0)^n}W(x_0)\sqrt{\cosh(x_0)}\sqrt{\sinh(x_0)}} - \frac{z\overline{z}\sqrt{\sinh(x_0)}}{4\sqrt{\epsilon}k\sqrt{W(x_0)^n}\cosh(x_0)^{\frac{3}{2}}} \\ &+ \frac{z^2\frac{\partial}{\partial x_0}W(x_0)}{4\sqrt{\epsilon}k\sqrt{W(x_0)^n}W(x_0)\sqrt{\cosh(x_0)}\sqrt{\sinh(x_0)}} - \frac{n\overline{z}\frac{\partial}{\partial x_0}W(x_0)}{8\sqrt{\epsilon}k\sqrt{W(x_0)^n}W(x_0)\sqrt{\cosh(x_0)}\sqrt{\sinh(x_0)}} + \frac{\overline{z}\sqrt{\cosh(x_0)}}{2\sqrt{\epsilon}k\sqrt{W(x_0)^n}\sinh(x_0)^{\frac{3}{2}}} \\ &- \frac{\overline{z}}{2\sqrt{\epsilon}k\sqrt{W(x_0)^n}\sqrt{\cosh(x_0)}\sqrt{\sinh(x_0)}} - \frac{\overline{z}\sqrt{\sinh(x_0)}}{4\sqrt{\epsilon}k\sqrt{W(x_0)^n}\cosh(x_0)^{\frac{3}{2}}} + \frac{n\frac{\partial}{\partial x_0}W(x_0)}{8\sqrt{\epsilon}k\sqrt{W(x_0)^n}W(x_0)\sqrt{\cosh(x_0)}\sqrt{\sinh(x_0)}} \\ &- \frac{z\overline{z}}{8\sqrt{\epsilon}k\sqrt{W(x_0)^n}\cosh(x_0)^{\frac{3}{2}}\sinh(x_0)^{\frac{3}{2}}} + \frac{\sqrt{\sinh(x_0)}}{4\sqrt{\epsilon}k\sqrt{W(x_0)^n}\cosh(x_0)^{\frac{3}{2}}} - \frac{3\overline{z}}{8\sqrt{\epsilon}k\sqrt{W(x_0)^n}\cosh(x_0)^{\frac{3}{2}}\sinh(x_0)^{\frac{3}{2}}} + \frac{1}{8\sqrt{\epsilon}k\sqrt{W(x_0)^n}\cosh(x_0)^{\frac{3}{2}}\sinh(x_0)^{\frac{3}{2}}}\end{aligned} \tag{81}$$

*Weyl components:*

$$\Psi_0' = \Psi_0 \tag{82}$$

$$\Psi_1' = -\frac{n^2 z W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)}{4\,\epsilon k^2 W(x_0)\cosh(x_0)\sinh(x_0)} - \frac{n z W(x_0)^{-n}}{2\,\epsilon k^2 \cosh(x_0)\sinh(x_0)} + \frac{n z W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)^2}{8\,\epsilon k^2 W(x_0)^2 \cosh(x_0)\sinh(x_0)}$$
$$+ \frac{z W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)}{4\,\epsilon k^2 W(x_0)\cosh(x_0)\sinh(x_0)} + \frac{z W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)^2}{8\,\epsilon k^2 W(x_0)^2 \cosh(x_0)\sinh(x_0)} - \frac{z W(x_0)^{-n} \frac{\partial^2}{(\partial x_0)^2} W(x_0)}{8\,\epsilon k^2 W(x_0)\cosh(x_0)\sinh(x_0)} \tag{83}$$

$$\Psi_2' = -\frac{n^2 z^2 W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)}{4\,\epsilon k^2 W(x_0)\cosh(x_0)\sinh(x_0)} - \frac{n z^2 W(x_0)^{-n}}{2\,\epsilon k^2 \cosh(x_0)\sinh(x_0)} + \frac{n z^2 W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)^2}{8\,\epsilon k^2 W(x_0)^2 \cosh(x_0)\sinh(x_0)} - \frac{n^2 W(x_0)^{-n}}{6\,\epsilon k^2 \cosh(x_0)\sinh(x_0)}$$
$$+ \frac{z^2 W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)}{4\,\epsilon k^2 W(x_0)\cosh(x_0)\sinh(x_0)} + \frac{z^2 W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)^2}{8\,\epsilon k^2 W(x_0)^2 \cosh(x_0)\sinh(x_0)} - \frac{z^2 W(x_0)^{-n} \frac{\partial^2}{(\partial x_0)^2} W(x_0)}{8\,\epsilon k^2 W(x_0)\cosh(x_0)\sinh(x_0)} \tag{84}$$
$$+ \frac{n W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)^2}{24\,\epsilon k^2 W(x_0)^2 \cosh(x_0)\sinh(x_0)} - \frac{n W(x_0)^{-n} \frac{\partial^2}{(\partial x_0)^2} W(x_0)}{24\,\epsilon k^2 W(x_0)\cosh(x_0)\sinh(x_0)} + \frac{W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)^2}{24\,\epsilon k^2 W(x_0)^2 \cosh(x_0)\sinh(x_0)}$$

$$\Psi_3' = -\frac{n^2 z^3 W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)}{4\,\epsilon k^2 W(x_0)\cosh(x_0)\sinh(x_0)} - \frac{n z^3 W(x_0)^{-n}}{2\,\epsilon k^2 \cosh(x_0)\sinh(x_0)} + \frac{n z^3 W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)^2}{8\,\epsilon k^2 W(x_0)^2 \cosh(x_0)\sinh(x_0)} - \frac{n^2 z W(x_0)^{-n}}{2\,\epsilon k^2 \cosh(x_0)\sinh(x_0)}$$
$$+ \frac{z^3 W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)}{4\,\epsilon k^2 W(x_0)\cosh(x_0)\sinh(x_0)} + \frac{z^3 W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)^2}{8\,\epsilon k^2 W(x_0)^2 \cosh(x_0)\sinh(x_0)} - \frac{z^3 W(x_0)^{-n} \frac{\partial^2}{(\partial x_0)^2} W(x_0)}{8\,\epsilon k^2 W(x_0)\cosh(x_0)\sinh(x_0)} \tag{85}$$
$$+ \frac{n z W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)^2}{8\,\epsilon k^2 W(x_0)^2 \cosh(x_0)\sinh(x_0)} - \frac{n z W(x_0)^{-n} \frac{\partial^2}{(\partial x_0)^2} W(x_0)}{8\,\epsilon k^2 W(x_0)\cosh(x_0)\sinh(x_0)} + \frac{z W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)^2}{8\,\epsilon k^2 W(x_0)^2 \cosh(x_0)\sinh(x_0)}$$

$$\Psi_4' = -\frac{n^2 z^4 W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)}{4\,\epsilon k^2 W(x_0)\cosh(x_0)\sinh(x_0)} - \frac{n z^4 W(x_0)^{-n}}{2\,\epsilon k^2 \cosh(x_0)\sinh(x_0)} + \frac{n z^4 W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)^2}{8\,\epsilon k^2 W(x_0)^2 \cosh(x_0)\sinh(x_0)} - \frac{n^2 z^2 W(x_0)^{-n}}{\epsilon k^2 \cosh(x_0)\sinh(x_0)}$$
$$+ \frac{z^4 W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)}{4\,\epsilon k^2 W(x_0)\cosh(x_0)\sinh(x_0)} + \frac{z^4 W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)^2}{8\,\epsilon k^2 W(x_0)^2 \cosh(x_0)\sinh(x_0)} - \frac{z^4 W(x_0)^{-n} \frac{\partial^2}{(\partial x_0)^2} W(x_0)}{8\,\epsilon k^2 W(x_0)\cosh(x_0)\sinh(x_0)} + \frac{n z^2 W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)^2}{4\,\epsilon k^2 W(x_0)^2 \cosh(x_0)\sinh(x_0)} \tag{86}$$
$$- \frac{n z^2 W(x_0)^{-n} \frac{\partial^2}{(\partial x_0)^2} W(x_0)}{4\,\epsilon k^2 W(x_0)\cosh(x_0)\sinh(x_0)} + \frac{n^2 W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)}{4\,\epsilon k^2 W(x_0)\cosh(x_0)\sinh(x_0)} + \frac{z^2 W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)^2}{4\,\epsilon k^2 W(x_0)^2 \cosh(x_0)\sinh(x_0)} - \frac{n W(x_0)^{-n}}{2\,\epsilon k^2 \cosh(x_0)\sinh(x_0)}$$
$$+ \frac{n W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)^2}{8\,\epsilon k^2 W(x_0)^2 \cosh(x_0)\sinh(x_0)} - \frac{W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)}{4\,\epsilon k^2 W(x_0)\cosh(x_0)\sinh(x_0)} + \frac{W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)^2}{8\,\epsilon k^2 W(x_0)^2 \cosh(x_0)\sinh(x_0)} - \frac{W(x_0)^{-n} \frac{\partial^2}{(\partial x_0)^2} W(x_0)}{8\,\epsilon k^2 W(x_0)\cosh(x_0)\sinh(x_0)}$$

*Ricci components:*

$$\Phi'_{00} = \Phi_{00} \tag{87}$$

$$\begin{aligned}\Phi'_{01} = &-\frac{n^2 z W(x_0)^{-n}\cosh(x_0)}{4\,\epsilon k^2 \sinh(x_0)^3} + \frac{n z W(x_0)^{-n}\frac{\partial}{\partial x_0}W(x_0)}{4\,\epsilon k^2 W(x_0)\sinh(x_0)^2} + \frac{n^2 z W(x_0)^{-n}}{4\,\epsilon k^2\cosh(x_0)\sinh(x_0)^3} - \frac{z W(x_0)^{-n}\cosh(x_0)\frac{\partial}{\partial x_0}W(x_0)^2}{16\,\epsilon k^2 W(x_0)^2\sinh(x_0)^3} \\ &- \frac{n z W(x_0)^{-n}\frac{\partial}{\partial x_0}W(x_0)}{8\,\epsilon k^2 W(x_0)\cosh(x_0)^2\sinh(x_0)^2} + \frac{z W(x_0)^{-n}\frac{\partial}{\partial x_0}W(x_0)^2}{16\,\epsilon k^2 W(x_0)^2\cosh(x_0)\sinh(x_0)^3} + \frac{3\,z W(x_0)^{-n}}{16\,\epsilon k^2\cosh(x_0)^3\sinh(x_0)^3}\end{aligned} \tag{88}$$

$$\begin{aligned}\Phi'_{02} = &-\frac{n^2 z^2 W(x_0)^{-n}\cosh(x_0)}{4\,\epsilon k^2 \sinh(x_0)^3} + \frac{n z^2 W(x_0)^{-n}\frac{\partial}{\partial x_0}W(x_0)}{4\,\epsilon k^2 W(x_0)\sinh(x_0)^2} + \frac{n^2 z^2 W(x_0)^{-n}}{4\,\epsilon k^2\cosh(x_0)\sinh(x_0)^3} - \frac{z^2 W(x_0)^{-n}\cosh(x_0)\frac{\partial}{\partial x_0}W(x_0)^2}{16\,\epsilon k^2 W(x_0)^2\sinh(x_0)^3} \\ &- \frac{n W(x_0)^{-n}}{2\,\epsilon k^2\cosh(x_0)\sinh(x_0)} - \frac{n z^2 W(x_0)^{-n}\frac{\partial}{\partial x_0}W(x_0)}{8\,\epsilon k^2 W(x_0)\cosh(x_0)^2\sinh(x_0)^2} + \frac{W(x_0)^{-n}\frac{\partial}{\partial x_0}W(x_0)}{4\,\epsilon k^2 W(x_0)\cosh(x_0)^2} + \frac{z^2 W(x_0)^{-n}\frac{\partial}{\partial x_0}W(x_0)^2}{16\,\epsilon k^2 W(x_0)^2\cosh(x_0)\sinh(x_0)^3} \\ &- \frac{W(x_0)^{-n}\frac{\partial}{\partial x_0}W(x_0)^2}{8\,\epsilon k^2 W(x_0)^2\cosh(x_0)\sinh(x_0)} + \frac{W(x_0)^{-n}\frac{\partial^2}{(\partial x_0)^2}W(x_0)}{8\,\epsilon k^2 W(x_0)\cosh(x_0)\sinh(x_0)} + \frac{3\,z^2 W(x_0)^{-n}}{16\,\epsilon k^2\cosh(x_0)^3\sinh(x_0)^3} + \frac{W(x_0)^{-n}\frac{\partial}{\partial x_0}W(x_0)}{8\,\epsilon k^2 W(x_0)\cosh(x_0)^2\sinh(x_0)^2}\end{aligned} \tag{89}$$

$$\begin{aligned}\Phi'_{10} = &-\frac{n^2 W(x_0)^{-n}\,\overline{z}\cosh(x_0)}{4\,\epsilon k^2 \sinh(x_0)^3} + \frac{n W(x_0)^{-n}\,\overline{z}\frac{\partial}{\partial x_0}W(x_0)}{4\,\epsilon k^2 W(x_0)\sinh(x_0)^2} + \frac{n^2 W(x_0)^{-n}\,\overline{z}}{4\,\epsilon k^2\cosh(x_0)\sinh(x_0)^3} - \frac{W(x_0)^{-n}\,\overline{z}\cosh(x_0)\frac{\partial}{\partial x_0}W(x_0)^2}{16\,\epsilon k^2 W(x_0)^2\sinh(x_0)^3} \\ &- \frac{n W(x_0)^{-n}\,\overline{z}\frac{\partial}{\partial x_0}W(x_0)}{8\,\epsilon k^2 W(x_0)\cosh(x_0)^2\sinh(x_0)^2} + \frac{W(x_0)^{-n}\,\overline{z}\frac{\partial}{\partial x_0}W(x_0)^2}{16\,\epsilon k^2 W(x_0)^2\cosh(x_0)\sinh(x_0)^3} + \frac{3\,W(x_0)^{-n}\,\overline{z}}{16\,\epsilon k^2\cosh(x_0)^3\sinh(x_0)^3}\end{aligned} \tag{90}$$

$$\begin{aligned}\Phi'_{11} = &-\frac{n^2 z W(x_0)^{-n}\,\overline{z}\cosh(x_0)}{4\,\epsilon k^2 \sinh(x_0)^3} + \frac{n z W(x_0)^{-n}\,\overline{z}\frac{\partial}{\partial x_0}W(x_0)}{4\,\epsilon k^2 W(x_0)\sinh(x_0)^2} + \frac{n^2 z W(x_0)^{-n}\,\overline{z}}{4\,\epsilon k^2\cosh(x_0)\sinh(x_0)^3} + \frac{n^2 W(x_0)^{-n}\cosh(x_0)}{8\,\epsilon k^2\sinh(x_0)^3} - \frac{z W(x_0)^{-n}\,\overline{z}\cosh(x_0)\frac{\partial}{\partial x_0}W(x_0)^2}{16\,\epsilon k^2 W(x_0)^2\sinh(x_0)^3} \\ &- \frac{n z W(x_0)^{-n}\,\overline{z}\frac{\partial}{\partial x_0}W(x_0)}{8\,\epsilon k^2 W(x_0)\cosh(x_0)^2\sinh(x_0)^2} + \frac{n W(x_0)^{-n}\cosh(x_0)\frac{\partial}{\partial x_0}W(x_0)^2}{16\,\epsilon k^2 W(x_0)^2\sinh(x_0)^3} - \frac{n W(x_0)^{-n}\cosh(x_0)\frac{\partial^2}{(\partial x_0)^2}W(x_0)}{16\,\epsilon k^2 W(x_0)\sinh(x_0)^3} \\ &- \frac{n^2 W(x_0)^{-n}}{8\,\epsilon k^2\cosh(x_0)\sinh(x_0)^3} + \frac{z W(x_0)^{-n}\,\overline{z}\frac{\partial}{\partial x_0}W(x_0)^2}{16\,\epsilon k^2 W(x_0)^2\cosh(x_0)\sinh(x_0)^3} - \frac{W(x_0)^{-n}\cosh(x_0)\frac{\partial}{\partial x_0}W(x_0)^2}{32\,\epsilon k^2 W(x_0)^2\sinh(x_0)^3} - \frac{n W(x_0)^{-n}\frac{\partial}{\partial x_0}W(x_0)^2}{16\,\epsilon k^2 W(x_0)^2\cosh(x_0)\sinh(x_0)^3} \\ &+ \frac{n W(x_0)^{-n}\frac{\partial^2}{(\partial x_0)^2}W(x_0)}{16\,\epsilon k^2 W(x_0)\cosh(x_0)\sinh(x_0)^3} + \frac{3\,z W(x_0)^{-n}\,\overline{z}}{16\,\epsilon k^2\cosh(x_0)^3\sinh(x_0)^3} + \frac{W(x_0)^{-n}\frac{\partial}{\partial x_0}W(x_0)^2}{32\,\epsilon k^2 W(x_0)^2\cosh(x_0)\sinh(x_0)^3} + \frac{3\,W(x_0)^{-n}}{32\,\epsilon k^2\cosh(x_0)^3\sinh(x_0)^3}\end{aligned} \tag{91}$$

$$
\begin{aligned}
\Phi'_{12} = & -\frac{n^2 z^2 W(x_0)^{-n}\,\overline{z}\cosh(x_0)}{4\,\epsilon k^2 \sinh(x_0)^3} + \frac{n z^2 W(x_0)^{-n}\,\overline{z}\,\frac{\partial}{\partial x_0} W(x_0)}{4\,\epsilon k^2 W(x_0)\sinh(x_0)^2} + \frac{n^2 z^2 W(x_0)^{-n}\,\overline{z}}{4\,\epsilon k^2 \cosh(x_0)\sinh(x_0)^3} + \frac{n^2 z W(x_0)^{-n}\cosh(x_0)}{4\,\epsilon k^2 \sinh(x_0)^3} - \frac{z^2 W(x_0)^{-n}\,\overline{z}\cosh(x_0)\,\frac{\partial}{\partial x_0} W(x_0)^2}{16\,\epsilon k^2 W(x_0)^2 \sinh(x_0)^3} \\
& - \frac{n W(x_0)^{-n}\,\overline{z}}{2\,\epsilon k^2 \cosh(x_0)\sinh(x_0)} - \frac{n z^2 W(x_0)^{-n}\,\overline{z}\,\frac{\partial}{\partial x_0} W(x_0)}{8\,\epsilon k^2 W(x_0)\cosh(x_0)^2 \sinh(x_0)^2} + \frac{n z W(x_0)^{-n}\cosh(x_0)\,\frac{\partial}{\partial x_0} W(x_0)^2}{8\,\epsilon k^2 W(x_0)^2 \sinh(x_0)^3} - \frac{n z W(x_0)^{-n}\cosh(x_0)\,\frac{\partial^2}{(\partial x_0)^2} W(x_0)}{8\,\epsilon k^2 W(x_0)\sinh(x_0)^3} \\
& - \frac{n^2 z W(x_0)^{-n}}{4\,\epsilon k^2 \cosh(x_0)\sinh(x_0)^3} + \frac{W(x_0)^{-n}\,\overline{z}\,\frac{\partial}{\partial x_0} W(x_0)}{4\,\epsilon k^2 W(x_0)\cosh(x_0)^2} + \frac{z^2 W(x_0)^{-n}\,\overline{z}\,\frac{\partial}{\partial x_0} W(x_0)^2}{16\,\epsilon k^2 W(x_0)^2 \cosh(x_0)\sinh(x_0)^3} - \frac{z W(x_0)^{-n}\cosh(x_0)\,\frac{\partial}{\partial x_0} W(x_0)^2}{16\,\epsilon k^2 W(x_0)^2 \sinh(x_0)^3} \\
& - \frac{W(x_0)^{-n}\,\overline{z}\,\frac{\partial}{\partial x_0} W(x_0)^2}{8\,\epsilon k^2 W(x_0)^2 \cosh(x_0)\sinh(x_0)} + \frac{W(x_0)^{-n}\,\overline{z}\,\frac{\partial^2}{(\partial x_0)^2} W(x_0)}{8\,\epsilon k^2 W(x_0)\cosh(x_0)\sinh(x_0)} - \frac{n z W(x_0)^{-n}\,\frac{\partial}{\partial x_0} W(x_0)^2}{8\,\epsilon k^2 W(x_0)^2 \cosh(x_0)\sinh(x_0)^3} + \frac{n z W(x_0)^{-n}\,\frac{\partial^2}{(\partial x_0)^2} W(x_0)}{8\,\epsilon k^2 W(x_0)\cosh(x_0)\sinh(x_0)^3} \\
& + \frac{3\,z^2 W(x_0)^{-n}\,\overline{z}}{16\,\epsilon k^2 \cosh(x_0)^3 \sinh(x_0)^3} + \frac{W(x_0)^{-n}\,\overline{z}\,\frac{\partial}{\partial x_0} W(x_0)}{8\,\epsilon k^2 W(x_0)\cosh(x_0)^2 \sinh(x_0)^2} + \frac{z W(x_0)^{-n}\,\frac{\partial}{\partial x_0} W(x_0)^2}{16\,\epsilon k^2 W(x_0)^2 \cosh(x_0)\sinh(x_0)^3} + \frac{3\,z W(x_0)^{-n}}{16\,\epsilon k^2 \cosh(x_0)^3 \sinh(x_0)^3}
\end{aligned}
\tag{92}
$$

$$
\begin{aligned}
\Phi'_{20} = & -\frac{n^2 W(x_0)^{-n}\,\overline{z}^2\cosh(x_0)}{4\,\epsilon k^2 \sinh(x_0)^3} + \frac{n W(x_0)^{-n}\,\overline{z}^2\,\frac{\partial}{\partial x_0} W(x_0)}{4\,\epsilon k^2 W(x_0)\sinh(x_0)^2} + \frac{n^2 W(x_0)^{-n}\,\overline{z}^2}{4\,\epsilon k^2 \cosh(x_0)\sinh(x_0)^3} - \frac{W(x_0)^{-n}\,\overline{z}^2\cosh(x_0)\,\frac{\partial}{\partial x_0} W(x_0)^2}{16\,\epsilon k^2 W(x_0)^2 \sinh(x_0)^3} \\
& - \frac{n W(x_0)^{-n}}{2\,\epsilon k^2 \cosh(x_0)\sinh(x_0)} - \frac{n W(x_0)^{-n}\,\overline{z}^2\,\frac{\partial}{\partial x_0} W(x_0)}{8\,\epsilon k^2 W(x_0)\cosh(x_0)^2 \sinh(x_0)^2} + \frac{W(x_0)^{-n}\,\frac{\partial}{\partial x_0} W(x_0)}{4\,\epsilon k^2 W(x_0)\cosh(x_0)^2} + \frac{W(x_0)^{-n}\,\overline{z}^2\,\frac{\partial}{\partial x_0} W(x_0)^2}{16\,\epsilon k^2 W(x_0)^2 \cosh(x_0)\sinh(x_0)^3} \\
& - \frac{W(x_0)^{-n}\,\frac{\partial}{\partial x_0} W(x_0)^2}{8\,\epsilon k^2 W(x_0)^2 \cosh(x_0)\sinh(x_0)} + \frac{W(x_0)^{-n}\,\frac{\partial^2}{(\partial x_0)^2} W(x_0)}{8\,\epsilon k^2 W(x_0)\cosh(x_0)\sinh(x_0)} + \frac{3\,W(x_0)^{-n}\,\overline{z}^2}{16\,\epsilon k^2 \cosh(x_0)^3 \sinh(x_0)^3} + \frac{W(x_0)^{-n}\,\frac{\partial}{\partial x_0} W(x_0)}{8\,\epsilon k^2 W(x_0)\cosh(x_0)^2 \sinh(x_0)^2}
\end{aligned}
\tag{93}
$$

$$
\begin{aligned}
\Phi'_{21} = & -\frac{n^2 z W(x_0)^{-n} \overline{z}^2 \cosh(x_0)}{4\,\epsilon k^2 \sinh(x_0)^3} + \frac{n z W(x_0)^{-n} \overline{z}^2 \frac{\partial}{\partial x_0} W(x_0)}{4\,\epsilon k^2 W(x_0) \sinh(x_0)^2} + \frac{n^2 z W(x_0)^{-n} \overline{z}^2}{4\,\epsilon k^2 \cosh(x_0) \sinh(x_0)^3} + \frac{n^2 W(x_0)^{-n} \overline{z} \cosh(x_0)}{4\,\epsilon k^2 \sinh(x_0)^3} - \frac{z W(x_0)^{-n} \overline{z}^2 \cosh(x_0) \frac{\partial}{\partial x_0} W(x_0)^2}{16\,\epsilon k^2 W(x_0)^2 \sinh(x_0)^3} \\
& - \frac{n z W(x_0)^{-n}}{2\,\epsilon k^2 \cosh(x_0) \sinh(x_0)} - \frac{n z W(x_0)^{-n} \overline{z}^2 \frac{\partial}{\partial x_0} W(x_0)}{8\,\epsilon k^2 W(x_0) \cosh(x_0)^2 \sinh(x_0)^2} + \frac{n W(x_0)^{-n} \overline{z} \cosh(x_0) \frac{\partial}{\partial x_0} W(x_0)^2}{8\,\epsilon k^2 W(x_0)^2 \sinh(x_0)^3} - \frac{n W(x_0)^{-n} \overline{z} \cosh(x_0) \frac{\partial^2}{(\partial x_0)^2} W(x_0)}{8\,\epsilon k^2 W(x_0) \sinh(x_0)^3} \\
& - \frac{n^2 W(x_0)^{-n} \overline{z}}{4\,\epsilon k^2 \cosh(x_0) \sinh(x_0)^3} + \frac{z W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)}{4\,\epsilon k^2 W(x_0) \cosh(x_0)^2} + \frac{z W(x_0)^{-n} \overline{z}^2 \frac{\partial}{\partial x_0} W(x_0)^2}{16\,\epsilon k^2 W(x_0)^2 \cosh(x_0) \sinh(x_0)^3} - \frac{W(x_0)^{-n} \overline{z} \cosh(x_0) \frac{\partial}{\partial x_0} W(x_0)^2}{16\,\epsilon k^2 W(x_0)^2 \sinh(x_0)^3} \\
& - \frac{z W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)^2}{8\,\epsilon k^2 W(x_0)^2 \cosh(x_0) \sinh(x_0)} + \frac{z W(x_0)^{-n} \frac{\partial^2}{(\partial x_0)^2} W(x_0)}{8\,\epsilon k^2 W(x_0) \cosh(x_0) \sinh(x_0)} - \frac{n W(x_0)^{-n} \overline{z} \frac{\partial}{\partial x_0} W(x_0)^2}{8\,\epsilon k^2 W(x_0)^2 \cosh(x_0) \sinh(x_0)^3} + \frac{n W(x_0)^{-n} \overline{z} \frac{\partial^2}{(\partial x_0)^2} W(x_0)}{8\,\epsilon k^2 W(x_0) \cosh(x_0) \sinh(x_0)^3} \\
& + \frac{3\, z W(x_0)^{-n} \overline{z}^2}{16\,\epsilon k^2 \cosh(x_0)^3 \sinh(x_0)^3} + \frac{z W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)}{8\,\epsilon k^2 W(x_0) \cosh(x_0)^2 \sinh(x_0)^2} + \frac{W(x_0)^{-n} \overline{z} \frac{\partial}{\partial x_0} W(x_0)^2}{16\,\epsilon k^2 W(x_0)^2 \cosh(x_0) \sinh(x_0)^3} + \frac{3\, W(x_0)^{-n} \overline{z}}{16\,\epsilon k^2 \cosh(x_0)^3 \sinh(x_0)^3}
\end{aligned}
\tag{94}
$$

$$
\begin{aligned}
\Phi'_{22} = & -\frac{n^2 z^2 W(x_0)^{-n} \overline{z}^2 \cosh(x_0)}{4\,\epsilon k^2 \sinh(x_0)^3} + \frac{n z^2 W(x_0)^{-n} \overline{z}^2 \frac{\partial}{\partial x_0} W(x_0)}{4\,\epsilon k^2 W(x_0) \sinh(x_0)^2} + \frac{n^2 z^2 W(x_0)^{-n} \overline{z}^2}{4\,\epsilon k^2 \cosh(x_0) \sinh(x_0)^3} + \frac{n^2 z W(x_0)^{-n} \overline{z} \cosh(x_0)}{2\,\epsilon k^2 \sinh(x_0)^3} - \frac{z^2 W(x_0)^{-n} \overline{z}^2 \cosh(x_0) \frac{\partial}{\partial x_0} W(x_0)^2}{16\,\epsilon k^2 W(x_0)^2 \sinh(x_0)^3} \\
& - \frac{n z^2 W(x_0)^{-n}}{2\,\epsilon k^2 \cosh(x_0) \sinh(x_0)} - \frac{n W(x_0)^{-n} \overline{z}^2}{2\,\epsilon k^2 \cosh(x_0) \sinh(x_0)} - \frac{n z^2 W(x_0)^{-n} \overline{z}^2 \frac{\partial}{\partial x_0} W(x_0)}{8\,\epsilon k^2 W(x_0) \cosh(x_0)^2 \sinh(x_0)^2} + \frac{n z W(x_0)^{-n} \overline{z} \cosh(x_0) \frac{\partial}{\partial x_0} W(x_0)^2}{4\,\epsilon k^2 W(x_0)^2 \sinh(x_0)^3} \\
& - \frac{n z W(x_0)^{-n} \overline{z} \cosh(x_0) \frac{\partial^2}{(\partial x_0)^2} W(x_0)}{4\,\epsilon k^2 W(x_0) \sinh(x_0)^3} - \frac{n^2 z W(x_0)^{-n} \overline{z}}{2\,\epsilon k^2 \cosh(x_0) \sinh(x_0)^3} - \frac{n^2 W(x_0)^{-n} \cosh(x_0)}{4\,\epsilon k^2 \sinh(x_0)^3} + \frac{z^2 W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)}{4\,\epsilon k^2 W(x_0) \cosh(x_0)^2} \\
& + \frac{W(x_0)^{-n} \overline{z}^2 \frac{\partial}{\partial x_0} W(x_0)}{4\,\epsilon k^2 W(x_0) \cosh(x_0)^2} + \frac{z^2 W(x_0)^{-n} \overline{z}^2 \frac{\partial}{\partial x_0} W(x_0)^2}{16\,\epsilon k^2 W(x_0)^2 \cosh(x_0) \sinh(x_0)^3} - \frac{z W(x_0)^{-n} \overline{z} \cosh(x_0) \frac{\partial}{\partial x_0} W(x_0)^2}{8\,\epsilon k^2 W(x_0)^2 \sinh(x_0)^3} - \frac{z^2 W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)^2}{8\,\epsilon k^2 W(x_0)^2 \cosh(x_0) \sinh(x_0)} \\
& - \frac{W(x_0)^{-n} \overline{z}^2 \frac{\partial}{\partial x_0} W(x_0)^2}{8\,\epsilon k^2 W(x_0)^2 \cosh(x_0) \sinh(x_0)} + \frac{z^2 W(x_0)^{-n} \frac{\partial^2}{(\partial x_0)^2} W(x_0)}{8\,\epsilon k^2 W(x_0) \cosh(x_0) \sinh(x_0)} + \frac{W(x_0)^{-n} \overline{z}^2 \frac{\partial^2}{(\partial x_0)^2} W(x_0)}{8\,\epsilon k^2 W(x_0) \cosh(x_0) \sinh(x_0)} + \frac{n W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)}{4\,\epsilon k^2 W(x_0) \sinh(x_0)^2} \\
& - \frac{n z W(x_0)^{-n} \overline{z} \frac{\partial}{\partial x_0} W(x_0)^2}{4\,\epsilon k^2 W(x_0)^2 \cosh(x_0) \sinh(x_0)^3} + \frac{n z W(x_0)^{-n} \overline{z} \frac{\partial^2}{(\partial x_0)^2} W(x_0)}{4\,\epsilon k^2 W(x_0) \cosh(x_0) \sinh(x_0)^3} + \frac{3\, z^2 W(x_0)^{-n} \overline{z}^2}{16\,\epsilon k^2 \cosh(x_0)^3 \sinh(x_0)^3} + \frac{n^2 W(x_0)^{-n}}{4\,\epsilon k^2 \cosh(x_0) \sinh(x_0)^3} \\
& + \frac{z^2 W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)}{8\,\epsilon k^2 W(x_0) \cosh(x_0)^2 \sinh(x_0)^2} + \frac{W(x_0)^{-n} \overline{z}^2 \frac{\partial}{\partial x_0} W(x_0)}{8\,\epsilon k^2 W(x_0) \cosh(x_0)^2 \sinh(x_0)^2} + \frac{z W(x_0)^{-n} \overline{z} \frac{\partial}{\partial x_0} W(x_0)^2}{8\,\epsilon k^2 W(x_0)^2 \cosh(x_0) \sinh(x_0)^3} - \frac{W(x_0)^{-n} \cosh(x_0) \frac{\partial}{\partial x_0} W(x_0)^2}{16\,\epsilon k^2 W(x_0)^2 \sinh(x_0)^3} \\
& - \frac{n W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)}{8\,\epsilon k^2 W(x_0) \cosh(x_0)^2 \sinh(x_0)^2} + \frac{3\, z W(x_0)^{-n} \overline{z}}{8\,\epsilon k^2 \cosh(x_0)^3 \sinh(x_0)^3} + \frac{W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)^2}{16\,\epsilon k^2 W(x_0)^2 \cosh(x_0) \sinh(x_0)^3} + \frac{3\, W(x_0)^{-n}}{16\,\epsilon k^2 \cosh(x_0)^3 \sinh(x_0)^3}
\end{aligned}
\tag{95}
$$

$$\Lambda' = \Lambda \tag{96}$$

B Type transformations:

*Spin coefficients:*

$$\rho' = \frac{z\overline{z}\sqrt{\cosh(x_0)}}{2\sqrt{\epsilon}k\sqrt{W(x_0)^n}\sinh(x_0)^{\frac{3}{2}}} - \frac{z\overline{z}}{2\sqrt{\epsilon}k\sqrt{W(x_0)^n}\sqrt{\cosh(x_0)}\sqrt{\sinh(x_0)}} - \frac{z\overline{z}}{4\sqrt{\epsilon}k\sqrt{W(x_0)^n}\cosh(x_0)^{\frac{3}{2}}\sinh(x_0)^{\frac{3}{2}}} \tag{97}$$

$$\sigma' = -\frac{nz^2}{2\sqrt{\epsilon}k\sqrt{W(x_0)^n}\sqrt{\cosh(x_0)}\sqrt{\sinh(x_0)}} + \frac{z^2\frac{\partial}{\partial x_0}W(x_0)}{4\sqrt{\epsilon}k\sqrt{W(x_0)^n}W(x_0)\sqrt{\cosh(x_0)}\sqrt{\sinh(x_0)}} \tag{98}$$

$$\kappa' = 0 \tag{99}$$

$$\mu' = -\frac{\sqrt{\cosh(x_0)}}{2\sqrt{\epsilon}kz\sqrt{W(x_0)^n}\overline{z}\sinh(x_0)^{\frac{3}{2}}} - \frac{1}{2\sqrt{\epsilon}kz\sqrt{W(x_0)^n}\overline{z}\sqrt{\cosh(x_0)}\sqrt{\sinh(x_0)}} + \frac{1}{4\sqrt{\epsilon}kz\sqrt{W(x_0)^n}\overline{z}\cosh(x_0)^{\frac{3}{2}}\sinh(x_0)^{\frac{3}{2}}} \tag{100}$$

$$\lambda' = -\frac{n}{2\sqrt{\epsilon}kz^2\sqrt{W(x_0)^n}\sqrt{\cosh(x_0)}\sqrt{\sinh(x_0)}} - \frac{\frac{\partial}{\partial x_0}W(x_0)}{4\sqrt{\epsilon}kz^2\sqrt{W(x_0)^n}W(x_0)\sqrt{\cosh(x_0)}\sqrt{\sinh(x_0)}} \tag{101}$$

$$\nu' = 0 \tag{102}$$

$$\alpha' = 0 \tag{103}$$

$$\beta' = 0 \tag{104}$$

$$\tau' = 0 \tag{105}$$

$$\pi' = 0 \tag{106}$$

$$\epsilon' = -\frac{n z \overline{z} \frac{\partial}{\partial x_0} W(x_0)}{8\sqrt{\epsilon} k \sqrt{W(x_0)^n} W(x_0) \sqrt{\cosh(x_0)} \sqrt{\sinh(x_0)}} - \frac{z\overline{z}\sqrt{\sinh(x_0)}}{4\sqrt{\epsilon} k \sqrt{W(x_0)^n} \cosh(x_0)^{\frac{3}{2}}} - \frac{z\overline{z}}{8\sqrt{\epsilon} k \sqrt{W(x_0)^n} \cosh(x_0)^{\frac{3}{2}} \sinh(x_0)^{\frac{3}{2}}} \tag{107}$$

$$\gamma' = \frac{n \frac{\partial}{\partial x_0} W(x_0)}{8\sqrt{\epsilon} k z \sqrt{W(x_0)^n} W(x_0) \overline{z} \sqrt{\cosh(x_0)} \sqrt{\sinh(x_0)}} + \frac{\sqrt{\sinh(x_0)}}{4\sqrt{\epsilon} k z \sqrt{W(x_0)^n} \overline{z} \cosh(x_0)^{\frac{3}{2}}} + \frac{1}{8\sqrt{\epsilon} k z \sqrt{W(x_0)^n} \overline{z} \cosh(x_0)^{\frac{3}{2}} \sinh(x_0)^{\frac{3}{2}}} \tag{108}$$

*Weyl components:*

$$\begin{aligned}\Psi_0' = &-\frac{n^2 z^4 W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)}{4\epsilon k^2 W(x_0) \cosh(x_0) \sinh(x_0)} - \frac{n z^4 W(x_0)^{-n}}{2\epsilon k^2 \cosh(x_0) \sinh(x_0)} + \frac{n z^4 W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)^2}{8\epsilon k^2 W(x_0)^2 \cosh(x_0) \sinh(x_0)} \\ &+ \frac{z^4 W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)}{4\epsilon k^2 W(x_0) \cosh(x_0) \sinh(x_0)} + \frac{z^4 W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)^2}{8\epsilon k^2 W(x_0)^2 \cosh(x_0) \sinh(x_0)} - \frac{z^4 W(x_0)^{-n} \frac{\partial^2}{(\partial x_0)^2} W(x_0)}{8\epsilon k^2 W(x_0) \cosh(x_0) \sinh(x_0)}\end{aligned} \tag{109}$$

$$\Psi_1' = 0 \tag{110}$$

$$\Psi_2' = -\frac{n^2 z^2 W(x_0)^{-n}}{6\epsilon k^2 \cosh(x_0) \sinh(x_0)} + \frac{n z^2 W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)^2}{24\epsilon k^2 W(x_0)^2 \cosh(x_0) \sinh(x_0)} - \frac{n z^2 W(x_0)^{-n} \frac{\partial^2}{(\partial x_0)^2} W(x_0)}{24\epsilon k^2 W(x_0) \cosh(x_0) \sinh(x_0)} + \frac{z^2 W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)^2}{24\epsilon k^2 W(x_0)^2 \cosh(x_0) \sinh(x_0)} \tag{111}$$

$$\Psi_3' = 0 \tag{112}$$

$$\Psi_4' = \Psi_4 \tag{113}$$

*Ricci components:*

$$\begin{aligned}\Phi_{00}' = &-\frac{n^2 z^2 W(x_0)^{-n} \overline{z}^2 \cosh(x_0)}{4\epsilon k^2 \sinh(x_0)^3} + \frac{n z^2 W(x_0)^{-n} \overline{z}^2 \frac{\partial}{\partial x_0} W(x_0)}{4\epsilon k^2 W(x_0) \sinh(x_0)^2} + \frac{n^2 z^2 W(x_0)^{-n} \overline{z}^2}{4\epsilon k^2 \cosh(x_0) \sinh(x_0)^3} - \frac{z^2 W(x_0)^{-n} \overline{z}^2 \cosh(x_0) \frac{\partial}{\partial x_0} W(x_0)^2}{16\epsilon k^2 W(x_0)^2 \sinh(x_0)^3} \\ &- \frac{n z^2 W(x_0)^{-n} \overline{z}^2 \frac{\partial}{\partial x_0} W(x_0)}{8\epsilon k^2 W(x_0) \cosh(x_0)^2 \sinh(x_0)^2} + \frac{z^2 W(x_0)^{-n} \overline{z}^2 \frac{\partial}{\partial x_0} W(x_0)^2}{16\epsilon k^2 W(x_0)^2 \cosh(x_0) \sinh(x_0)^3} + \frac{3 z^2 W(x_0)^{-n} \overline{z}^2}{16\epsilon k^2 \cosh(x_0)^3 \sinh(x_0)^3}\end{aligned} \tag{114}$$

$$\Phi'_{01} = 0 \tag{115}$$

$$\begin{aligned}\Phi'_{02} = &-\frac{nz^2W\left(x_0\right)^{-n}}{2\,\epsilon k^2\cosh\left(x_0\right)\sinh\left(x_0\right)} + \frac{z^2W\left(x_0\right)^{-n}\frac{\partial}{\partial x_0}W\left(x_0\right)}{4\,\epsilon k^2W\left(x_0\right)\cosh\left(x_0\right)^2} - \frac{z^2W\left(x_0\right)^{-n}\frac{\partial}{\partial x_0}W\left(x_0\right)^2}{8\,\epsilon k^2W\left(x_0\right)^2\cosh\left(x_0\right)\sinh\left(x_0\right)} \\ &+ \frac{z^2W\left(x_0\right)^{-n}\frac{\partial^2}{(\partial x_0)^2}W\left(x_0\right)}{8\,\epsilon k^2W\left(x_0\right)\cosh\left(x_0\right)\sinh\left(x_0\right)} + \frac{z^2W\left(x_0\right)^{-n}\frac{\partial}{\partial x_0}W\left(x_0\right)}{8\,\epsilon k^2W\left(x_0\right)\cosh\left(x_0\right)^2\sinh\left(x_0\right)^2}\end{aligned} \tag{116}$$

$$\Phi'_{10} = 0 \tag{117}$$

$$\begin{aligned}\Phi'_{11} = &\frac{n^2zW\left(x_0\right)^{-n}\overline{z}\cosh\left(x_0\right)}{8\,\epsilon k^2\sinh\left(x_0\right)^3} + \frac{nzW\left(x_0\right)^{-n}\overline{z}\cosh\left(x_0\right)\frac{\partial}{\partial x_0}W\left(x_0\right)^2}{16\,\epsilon k^2W\left(x_0\right)^2\sinh\left(x_0\right)^3} - \frac{nzW\left(x_0\right)^{-n}\overline{z}\cosh\left(x_0\right)\frac{\partial^2}{(\partial x_0)^2}W\left(x_0\right)}{16\,\epsilon k^2W\left(x_0\right)\sinh\left(x_0\right)^3} \\ &- \frac{n^2zW\left(x_0\right)^{-n}\overline{z}}{8\,\epsilon k^2\cosh\left(x_0\right)\sinh\left(x_0\right)^3} - \frac{zW\left(x_0\right)^{-n}\overline{z}\cosh\left(x_0\right)\frac{\partial}{\partial x_0}W\left(x_0\right)^2}{32\,\epsilon k^2W\left(x_0\right)^2\sinh\left(x_0\right)^3} - \frac{nzW\left(x_0\right)^{-n}\overline{z}\frac{\partial}{\partial x_0}W\left(x_0\right)^2}{16\,\epsilon k^2W\left(x_0\right)^2\cosh\left(x_0\right)\sinh\left(x_0\right)^3} \\ &+ \frac{nzW\left(x_0\right)^{-n}\overline{z}\frac{\partial^2}{(\partial x_0)^2}W\left(x_0\right)}{16\,\epsilon k^2W\left(x_0\right)\cosh\left(x_0\right)\sinh\left(x_0\right)^3} + \frac{zW\left(x_0\right)^{-n}\overline{z}\frac{\partial}{\partial x_0}W\left(x_0\right)^2}{32\,\epsilon k^2W\left(x_0\right)^2\cosh\left(x_0\right)\sinh\left(x_0\right)^3} + \frac{3\,zW\left(x_0\right)^{-n}\overline{z}}{32\,\epsilon k^2\cosh\left(x_0\right)^3\sinh\left(x_0\right)^3}\end{aligned} \tag{118}$$

$$\Phi'_{12} = 0 \tag{119}$$

$$\begin{aligned}\Phi'_{20} = &-\frac{nW\left(x_0\right)^{-n}\overline{z}^2}{2\,\epsilon k^2\cosh\left(x_0\right)\sinh\left(x_0\right)} + \frac{W\left(x_0\right)^{-n}\overline{z}^2\frac{\partial}{\partial x_0}W\left(x_0\right)}{4\,\epsilon k^2W\left(x_0\right)\cosh\left(x_0\right)^2} - \frac{W\left(x_0\right)^{-n}\overline{z}^2\frac{\partial}{\partial x_0}W\left(x_0\right)^2}{8\,\epsilon k^2W\left(x_0\right)^2\cosh\left(x_0\right)\sinh\left(x_0\right)} \\ &+ \frac{W\left(x_0\right)^{-n}\overline{z}^2\frac{\partial^2}{(\partial x_0)^2}W\left(x_0\right)}{8\,\epsilon k^2W\left(x_0\right)\cosh\left(x_0\right)\sinh\left(x_0\right)} + \frac{W\left(x_0\right)^{-n}\overline{z}^2\frac{\partial}{\partial x_0}W\left(x_0\right)}{8\,\epsilon k^2W\left(x_0\right)\cosh\left(x_0\right)^2\sinh\left(x_0\right)^2}\end{aligned} \tag{120}$$

$$\Phi'_{21} = 0 \tag{121}$$

$$\Phi'_{22} = \Phi_{22} \tag{122}$$

$$\Lambda' = \Lambda \tag{123}$$

C Type transformations:

*Spin coefficients:*

$$\begin{aligned}\rho' = &-\frac{z\sqrt{\cosh(x_0)}}{2\sqrt{\epsilon}k\sqrt{W(x_0)^n}\sinh(x_0)^{\frac{3}{2}}} - \frac{\overline{z}\sqrt{\cosh(x_0)}}{2\sqrt{\epsilon}k\sqrt{W(x_0)^n}\sinh(x_0)^{\frac{3}{2}}} - \frac{z}{2\sqrt{\epsilon}k\sqrt{W(x_0)^n}\sqrt{\cosh(x_0)}\sqrt{\sinh(x_0)}} \\ &- \frac{\overline{z}}{2\sqrt{\epsilon}k\sqrt{W(x_0)^n}\sqrt{\cosh(x_0)}\sqrt{\sinh(x_0)}} + \frac{\sqrt{\cosh(x_0)}}{2\sqrt{\epsilon}k\sqrt{W(x_0)^n}\sinh(x_0)^{\frac{3}{2}}} - \frac{1}{2\sqrt{\epsilon}k\sqrt{W(x_0)^n}\sqrt{\cosh(x_0)}\sqrt{\sinh(x_0)}} \\ &+ \frac{z}{4\sqrt{\epsilon}k\sqrt{W(x_0)^n}\cosh(x_0)^{\frac{3}{2}}\sinh(x_0)^{\frac{3}{2}}} + \frac{\overline{z}}{4\sqrt{\epsilon}k\sqrt{W(x_0)^n}\cosh(x_0)^{\frac{3}{2}}\sinh(x_0)^{\frac{3}{2}}} - \frac{1}{4\sqrt{\epsilon}k\sqrt{W(x_0)^n}\cosh(x_0)^{\frac{3}{2}}\sinh(x_0)^{\frac{3}{2}}}\end{aligned} \tag{124}$$

$$\begin{aligned}\sigma' = &\frac{nz^2\frac{\partial}{\partial x_0}W(x_0)}{4\sqrt{\epsilon}k\sqrt{W(x_0)^n}W(x_0)\sqrt{\cosh(x_0)}\sqrt{\sinh(x_0)}} - \frac{z^2\sqrt{\cosh(x_0)}}{2\sqrt{\epsilon}k\sqrt{W(x_0)^n}\sinh(x_0)^{\frac{3}{2}}} - \frac{z^2}{2\sqrt{\epsilon}k\sqrt{W(x_0)^n}\sqrt{\cosh(x_0)}\sqrt{\sinh(x_0)}} + \frac{z^2\sqrt{\sinh(x_0)}}{2\sqrt{\epsilon}k\sqrt{W(x_0)^n}\cosh(x_0)^{\frac{3}{2}}} \\ &- \frac{n}{2\sqrt{\epsilon}k\sqrt{W(x_0)^n}\sqrt{\cosh(x_0)}\sqrt{\sinh(x_0)}} + \frac{z^2}{2\sqrt{\epsilon}k\sqrt{W(x_0)^n}\cosh(x_0)^{\frac{3}{2}}\sinh(x_0)^{\frac{3}{2}}} + \frac{\frac{\partial}{\partial x_0}W(x_0)}{4\sqrt{\epsilon}k\sqrt{W(x_0)^n}W(x_0)\sqrt{\cosh(x_0)}\sqrt{\sinh(x_0)}}\end{aligned} \tag{125}$$

$$\begin{aligned}\kappa' = &-\frac{nz^3}{2\sqrt{\epsilon}k\sqrt{W(x_0)^n}\sqrt{\cosh(x_0)}\sqrt{\sinh(x_0)}} + \frac{nz^2\overline{z}\frac{\partial}{\partial x_0}W(x_0)}{4\sqrt{\epsilon}k\sqrt{W(x_0)^n}W(x_0)\sqrt{\cosh(x_0)}\sqrt{\sinh(x_0)}} - \frac{z^2\overline{z}\sqrt{\cosh(x_0)}}{2\sqrt{\epsilon}k\sqrt{W(x_0)^n}\sinh(x_0)^{\frac{3}{2}}} \\ &- \frac{z^2\overline{z}}{2\sqrt{\epsilon}k\sqrt{W(x_0)^n}\sqrt{\cosh(x_0)}\sqrt{\sinh(x_0)}} + \frac{z^2\overline{z}\sqrt{\sinh(x_0)}}{2\sqrt{\epsilon}k\sqrt{W(x_0)^n}\cosh(x_0)^{\frac{3}{2}}} - \frac{z^3\frac{\partial}{\partial x_0}W(x_0)}{4\sqrt{\epsilon}k\sqrt{W(x_0)^n}W(x_0)\sqrt{\cosh(x_0)}\sqrt{\sinh(x_0)}} \\ &- \frac{nz}{2\sqrt{\epsilon}k\sqrt{W(x_0)^n}\sqrt{\cosh(x_0)}\sqrt{\sinh(x_0)}} - \frac{nz\frac{\partial}{\partial x_0}W(x_0)}{4\sqrt{\epsilon}k\sqrt{W(x_0)^n}W(x_0)\sqrt{\cosh(x_0)}\sqrt{\sinh(x_0)}} + \frac{z^2\overline{z}}{2\sqrt{\epsilon}k\sqrt{W(x_0)^n}\cosh(x_0)^{\frac{3}{2}}\sinh(x_0)^{\frac{3}{2}}} \\ &- \frac{z\sqrt{\sinh(x_0)}}{2\sqrt{\epsilon}k\sqrt{W(x_0)^n}\cosh(x_0)^{\frac{3}{2}}} + \frac{z\frac{\partial}{\partial x_0}W(x_0)}{4\sqrt{\epsilon}k\sqrt{W(x_0)^n}W(x_0)\sqrt{\cosh(x_0)}\sqrt{\sinh(x_0)}} - \frac{z}{4\sqrt{\epsilon}k\sqrt{W(x_0)^n}\cosh(x_0)^{\frac{3}{2}}\sinh(x_0)^{\frac{3}{2}}}\end{aligned} \tag{126}$$

$$\mu' = -\frac{nz^2}{2\sqrt{\epsilon}k\sqrt{W(x_0)^n}\sqrt{\cosh(x_0)}\sqrt{\sinh(x_0)}} + \frac{nz\overline{z}\frac{\partial}{\partial x_0}W(x_0)}{4\sqrt{\epsilon}k\sqrt{W(x_0)^n}W(x_0)\sqrt{\cosh(x_0)}\sqrt{\sinh(x_0)}}$$
$$+ \frac{z\overline{z}\sqrt{\sinh(x_0)}}{2\sqrt{\epsilon}k\sqrt{W(x_0)^n}\cosh(x_0)^{\frac{3}{2}}} - \frac{z^2\frac{\partial}{\partial x_0}W(x_0)}{4\sqrt{\epsilon}k\sqrt{W(x_0)^n}W(x_0)\sqrt{\cosh(x_0)}\sqrt{\sinh(x_0)}} + \frac{z\overline{z}}{4\sqrt{\epsilon}k\sqrt{W(x_0)^n}\cosh(x_0)^{\frac{3}{2}}\sinh(x_0)^{\frac{3}{2}}}$$
$$- \frac{\sqrt{\cosh(x_0)}}{2\sqrt{\epsilon}k\sqrt{W(x_0)^n}\sinh(x_0)^{\frac{3}{2}}} - \frac{1}{2\sqrt{\epsilon}k\sqrt{W(x_0)^n}\sqrt{\cosh(x_0)}\sqrt{\sinh(x_0)}} + \frac{1}{4\sqrt{\epsilon}k\sqrt{W(x_0)^n}\cosh(x_0)^{\frac{3}{2}}\sinh(x_0)^{\frac{3}{2}}} \tag{127}$$

$$\lambda' = \lambda \tag{128}$$

$$\nu' = 0 \tag{129}$$

$$\alpha' = \frac{nz\frac{\partial}{\partial x_0}W(x_0)}{8\sqrt{\epsilon}k\sqrt{W(x_0)^n}W(x_0)\sqrt{\cosh(x_0)}\sqrt{\sinh(x_0)}} + \frac{z\sqrt{\sinh(x_0)}}{4\sqrt{\epsilon}k\sqrt{W(x_0)^n}\cosh(x_0)^{\frac{3}{2}}} + \frac{z}{8\sqrt{\epsilon}k\sqrt{W(x_0)^n}\cosh(x_0)^{\frac{3}{2}}\sinh(x_0)^{\frac{3}{2}}} \tag{130}$$

$$\beta' = \frac{nz\frac{\partial}{\partial x_0}W(x_0)}{8\sqrt{\epsilon}k\sqrt{W(x_0)^n}W(x_0)\sqrt{\cosh(x_0)}\sqrt{\sinh(x_0)}} - \frac{z\sqrt{\cosh(x_0)}}{2\sqrt{\epsilon}k\sqrt{W(x_0)^n}\sinh(x_0)^{\frac{3}{2}}}$$
$$- \frac{z}{2\sqrt{\epsilon}k\sqrt{W(x_0)^n}\sqrt{\cosh(x_0)}\sqrt{\sinh(x_0)}} + \frac{z\sqrt{\sinh(x_0)}}{4\sqrt{\epsilon}k\sqrt{W(x_0)^n}\cosh(x_0)^{\frac{3}{2}}} + \frac{3z}{8\sqrt{\epsilon}k\sqrt{W(x_0)^n}\cosh(x_0)^{\frac{3}{2}}\sinh(x_0)^{\frac{3}{2}}} \tag{131}$$

$$\tau' = \frac{nz\frac{\partial}{\partial x_0}W(x_0)}{4\sqrt{\epsilon}k\sqrt{W(x_0)^n}W(x_0)\sqrt{\cosh(x_0)}\sqrt{\sinh(x_0)}} + \frac{z\sqrt{\sinh(x_0)}}{2\sqrt{\epsilon}k\sqrt{W(x_0)^n}\cosh(x_0)^{\frac{3}{2}}} + \frac{z}{4\sqrt{\epsilon}k\sqrt{W(x_0)^n}\cosh(x_0)^{\frac{3}{2}}\sinh(x_0)^{\frac{3}{2}}} \tag{132}$$

$$\pi' = 0 \tag{133}$$

$$
\begin{aligned}
\epsilon' = & -\frac{nz^2}{2\sqrt{\epsilon}k\sqrt{W(x_0)^n}\sqrt{\cosh(x_0)}\sqrt{\sinh(x_0)}} + \frac{nz\overline{z}\frac{\partial}{\partial x_0}W(x_0)}{8\sqrt{\epsilon}k\sqrt{W(x_0)^n}W(x_0)\sqrt{\cosh(x_0)}\sqrt{\sinh(x_0)}} \\
& - \frac{z\overline{z}\sqrt{\cosh(x_0)}}{2\sqrt{\epsilon}k\sqrt{W(x_0)^n}\sinh(x_0)^{\frac{3}{2}}} - \frac{z\overline{z}}{2\sqrt{\epsilon}k\sqrt{W(x_0)^n}\sqrt{\cosh(x_0)}\sqrt{\sinh(x_0)}} + \frac{z\overline{z}\sqrt{\sinh(x_0)}}{4\sqrt{\epsilon}k\sqrt{W(x_0)^n}\cosh(x_0)^{\frac{3}{2}}} \\
& - \frac{z^2\frac{\partial}{\partial x_0}W(x_0)}{4\sqrt{\epsilon}k\sqrt{W(x_0)^n}W(x_0)\sqrt{\cosh(x_0)}\sqrt{\sinh(x_0)}} - \frac{n\frac{\partial}{\partial x_0}W(x_0)}{8\sqrt{\epsilon}k\sqrt{W(x_0)^n}W(x_0)\sqrt{\cosh(x_0)}\sqrt{\sinh(x_0)}} \\
& + \frac{3\,z\overline{z}}{8\sqrt{\epsilon}k\sqrt{W(x_0)^n}\cosh(x_0)^{\frac{3}{2}}\sinh(x_0)^{\frac{3}{2}}} - \frac{\sqrt{\sinh(x_0)}}{4\sqrt{\epsilon}k\sqrt{W(x_0)^n}\cosh(x_0)^{\frac{3}{2}}} - \frac{1}{8\sqrt{\epsilon}k\sqrt{W(x_0)^n}\cosh(x_0)^{\frac{3}{2}}\sinh(x_0)^{\frac{3}{2}}}
\end{aligned} \tag{134}
$$

$$
\gamma' = \gamma \tag{135}
$$

*Weyl components:*

$$
\begin{aligned}
\Psi_0' = & \frac{n^2z^4W(x_0)^{-n}\frac{\partial}{\partial x_0}W(x_0)}{4\,\epsilon k^2W(x_0)\cosh(x_0)\sinh(x_0)} - \frac{nz^4W(x_0)^{-n}}{2\,\epsilon k^2\cosh(x_0)\sinh(x_0)} + \frac{nz^4W(x_0)^{-n}\frac{\partial}{\partial x_0}W(x_0)^2}{8\,\epsilon k^2W(x_0)^2\cosh(x_0)\sinh(x_0)} - \frac{n^2z^2W(x_0)^{-n}}{\epsilon k^2\cosh(x_0)\sinh(x_0)} \\
& - \frac{z^4W(x_0)^{-n}\frac{\partial}{\partial x_0}W(x_0)}{4\,\epsilon k^2W(x_0)\cosh(x_0)\sinh(x_0)} + \frac{z^4W(x_0)^{-n}\frac{\partial}{\partial x_0}W(x_0)^2}{8\,\epsilon k^2W(x_0)^2\cosh(x_0)\sinh(x_0)} - \frac{z^4W(x_0)^{-n}\frac{\partial^2}{(\partial x_0)^2}W(x_0)}{8\,\epsilon k^2W(x_0)\cosh(x_0)\sinh(x_0)} + \frac{nz^2W(x_0)^{-n}\frac{\partial}{\partial x_0}W(x_0)^2}{4\,\epsilon k^2W(x_0)^2\cosh(x_0)\sinh(x_0)} \\
& - \frac{nz^2W(x_0)^{-n}\frac{\partial^2}{(\partial x_0)^2}W(x_0)}{4\,\epsilon k^2W(x_0)\cosh(x_0)\sinh(x_0)} - \frac{n^2W(x_0)^{-n}\frac{\partial}{\partial x_0}W(x_0)}{4\,\epsilon k^2W(x_0)\cosh(x_0)\sinh(x_0)} + \frac{z^2W(x_0)^{-n}\frac{\partial}{\partial x_0}W(x_0)^2}{4\,\epsilon k^2W(x_0)^2\cosh(x_0)\sinh(x_0)} - \frac{nW(x_0)^{-n}}{2\,\epsilon k^2\cosh(x_0)\sinh(x_0)} \\
& + \frac{nW(x_0)^{-n}\frac{\partial}{\partial x_0}W(x_0)^2}{8\,\epsilon k^2W(x_0)^2\cosh(x_0)\sinh(x_0)} + \frac{W(x_0)^{-n}\frac{\partial}{\partial x_0}W(x_0)}{4\,\epsilon k^2W(x_0)\cosh(x_0)\sinh(x_0)} + \frac{W(x_0)^{-n}\frac{\partial}{\partial x_0}W(x_0)^2}{8\,\epsilon k^2W(x_0)^2\cosh(x_0)\sinh(x_0)} - \frac{W(x_0)^{-n}\frac{\partial^2}{(\partial x_0)^2}W(x_0)}{8\,\epsilon k^2W(x_0)\cosh(x_0)\sinh(x_0)}
\end{aligned} \tag{136}
$$

$$
\begin{aligned}
\Psi_1' = & \frac{n^2z^3W(x_0)^{-n}\frac{\partial}{\partial x_0}W(x_0)}{4\,\epsilon k^2W(x_0)\cosh(x_0)\sinh(x_0)} - \frac{nz^3W(x_0)^{-n}}{2\,\epsilon k^2\cosh(x_0)\sinh(x_0)} + \frac{nz^3W(x_0)^{-n}\frac{\partial}{\partial x_0}W(x_0)^2}{8\,\epsilon k^2W(x_0)^2\cosh(x_0)\sinh(x_0)} - \frac{n^2zW(x_0)^{-n}}{2\,\epsilon k^2\cosh(x_0)\sinh(x_0)} \\
& - \frac{z^3W(x_0)^{-n}\frac{\partial}{\partial x_0}W(x_0)}{4\,\epsilon k^2W(x_0)\cosh(x_0)\sinh(x_0)} + \frac{z^3W(x_0)^{-n}\frac{\partial}{\partial x_0}W(x_0)^2}{8\,\epsilon k^2W(x_0)^2\cosh(x_0)\sinh(x_0)} - \frac{z^3W(x_0)^{-n}\frac{\partial^2}{(\partial x_0)^2}W(x_0)}{8\,\epsilon k^2W(x_0)\cosh(x_0)\sinh(x_0)} \\
& + \frac{nzW(x_0)^{-n}\frac{\partial}{\partial x_0}W(x_0)^2}{8\,\epsilon k^2W(x_0)^2\cosh(x_0)\sinh(x_0)} - \frac{nzW(x_0)^{-n}\frac{\partial^2}{(\partial x_0)^2}W(x_0)}{8\,\epsilon k^2W(x_0)\cosh(x_0)\sinh(x_0)} + \frac{zW(x_0)^{-n}\frac{\partial}{\partial x_0}W(x_0)^2}{8\,\epsilon k^2W(x_0)^2\cosh(x_0)\sinh(x_0)}
\end{aligned} \tag{137}
$$

$$\Psi_2' = \frac{n^2 z^2 W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)}{4\,\epsilon k^2 W(x_0)\cosh(x_0)\sinh(x_0)} - \frac{n z^2 W(x_0)^{-n}}{2\,\epsilon k^2 \cosh(x_0)\sinh(x_0)} + \frac{n z^2 W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)^2}{8\,\epsilon k^2 W(x_0)^2 \cosh(x_0)\sinh(x_0)} - \frac{n^2 W(x_0)^{-n}}{6\,\epsilon k^2 \cosh(x_0)\sinh(x_0)}$$
$$- \frac{z^2 W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)}{4\,\epsilon k^2 W(x_0)\cosh(x_0)\sinh(x_0)} + \frac{z^2 W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)^2}{8\,\epsilon k^2 W(x_0)^2 \cosh(x_0)\sinh(x_0)} - \frac{z^2 W(x_0)^{-n} \frac{\partial^2}{(\partial x_0)^2} W(x_0)}{8\,\epsilon k^2 W(x_0)\cosh(x_0)\sinh(x_0)} \tag{138}$$
$$+ \frac{n W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)^2}{24\,\epsilon k^2 W(x_0)^2 \cosh(x_0)\sinh(x_0)} - \frac{n W(x_0)^{-n} \frac{\partial^2}{(\partial x_0)^2} W(x_0)}{24\,\epsilon k^2 W(x_0)\cosh(x_0)\sinh(x_0)} + \frac{W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)^2}{24\,\epsilon k^2 W(x_0)^2 \cosh(x_0)\sinh(x_0)}$$

$$\Psi_3' = \frac{n^2 z W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)}{4\,\epsilon k^2 W(x_0)\cosh(x_0)\sinh(x_0)} - \frac{n z W(x_0)^{-n}}{2\,\epsilon k^2 \cosh(x_0)\sinh(x_0)} + \frac{n z W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)^2}{8\,\epsilon k^2 W(x_0)^2 \cosh(x_0)\sinh(x_0)}$$
$$- \frac{z W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)}{4\,\epsilon k^2 W(x_0)\cosh(x_0)\sinh(x_0)} + \frac{z W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)^2}{8\,\epsilon k^2 W(x_0)^2 \cosh(x_0)\sinh(x_0)} - \frac{z W(x_0)^{-n} \frac{\partial^2}{(\partial x_0)^2} W(x_0)}{8\,\epsilon k^2 W(x_0)\cosh(x_0)\sinh(x_0)} \tag{139}$$

$$\Psi_4' = \Psi_4 \tag{140}$$

*Ricci components:*

$$
\begin{aligned}
\Phi'_{00} = & -\frac{n^2 z^2 W(x_0)^{-n} \overline{z}^2 \cosh(x_0)}{4\,\epsilon k^2 \sinh(x_0)^3} + \frac{n z^2 W(x_0)^{-n} \overline{z}^2 \frac{\partial}{\partial x_0} W(x_0)}{4\,\epsilon k^2 W(x_0) \sinh(x_0)^2} + \frac{n^2 z^2 W(x_0)^{-n} \overline{z}^2}{4\,\epsilon k^2 \cosh(x_0) \sinh(x_0)^3} + \frac{n^2 z W(x_0)^{-n} \overline{z} \cosh(x_0)}{2\,\epsilon k^2 \sinh(x_0)^3} - \frac{z^2 W(x_0)^{-n} \overline{z}^2 \cosh(x_0) \frac{\partial}{\partial x_0} W(x_0)^2}{16\,\epsilon k^2 W(x_0)^2 \sinh(x_0)^3} \\
& - \frac{n z^2 W(x_0)^{-n}}{2\,\epsilon k^2 \cosh(x_0) \sinh(x_0)} - \frac{n W(x_0)^{-n} \overline{z}^2}{2\,\epsilon k^2 \cosh(x_0) \sinh(x_0)} - \frac{n z^2 W(x_0)^{-n} \overline{z}^2 \frac{\partial}{\partial x_0} W(x_0)}{8\,\epsilon k^2 W(x_0) \cosh(x_0)^2 \sinh(x_0)^2} + \frac{n z W(x_0)^{-n} \overline{z} \cosh(x_0) \frac{\partial}{\partial x_0} W(x_0)^2}{4\,\epsilon k^2 W(x_0)^2 \sinh(x_0)^3} \\
& - \frac{n z W(x_0)^{-n} \overline{z} \cosh(x_0) \frac{\partial^2}{(\partial x_0)^2} W(x_0)}{4\,\epsilon k^2 W(x_0) \sinh(x_0)^3} - \frac{n^2 z W(x_0)^{-n} \overline{z}}{2\,\epsilon k^2 \cosh(x_0) \sinh(x_0)^3} - \frac{n^2 W(x_0)^{-n} \cosh(x_0)}{4\,\epsilon k^2 \sinh(x_0)^3} + \frac{z^2 W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)}{4\,\epsilon k^2 W(x_0) \cosh(x_0)^2} \\
& + \frac{W(x_0)^{-n} \overline{z}^2 \frac{\partial}{\partial x_0} W(x_0)}{4\,\epsilon k^2 W(x_0) \cosh(x_0)^2} + \frac{z^2 W(x_0)^{-n} \overline{z}^2 \frac{\partial}{\partial x_0} W(x_0)^2}{16\,\epsilon k^2 W(x_0)^2 \cosh(x_0) \sinh(x_0)^3} - \frac{z W(x_0)^{-n} \overline{z} \cosh(x_0) \frac{\partial}{\partial x_0} W(x_0)^2}{8\,\epsilon k^2 W(x_0)^2 \sinh(x_0)^3} - \frac{z^2 W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)^2}{8\,\epsilon k^2 W(x_0)^2 \cosh(x_0) \sinh(x_0)} \\
& - \frac{W(x_0)^{-n} \overline{z}^2 \frac{\partial}{\partial x_0} W(x_0)^2}{8\,\epsilon k^2 W(x_0)^2 \cosh(x_0) \sinh(x_0)} + \frac{z^2 W(x_0)^{-n} \frac{\partial^2}{(\partial x_0)^2} W(x_0)}{8\,\epsilon k^2 W(x_0) \cosh(x_0) \sinh(x_0)} + \frac{W(x_0)^{-n} \overline{z}^2 \frac{\partial^2}{(\partial x_0)^2} W(x_0)}{8\,\epsilon k^2 W(x_0) \cosh(x_0) \sinh(x_0)} + \frac{n W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)}{4\,\epsilon k^2 W(x_0) \sinh(x_0)^2} \\
& - \frac{n z W(x_0)^{-n} \overline{z} \frac{\partial}{\partial x_0} W(x_0)^2}{4\,\epsilon k^2 W(x_0)^2 \cosh(x_0) \sinh(x_0)^3} + \frac{n z W(x_0)^{-n} \overline{z} \frac{\partial^2}{(\partial x_0)^2} W(x_0)}{4\,\epsilon k^2 W(x_0) \cosh(x_0) \sinh(x_0)^3} + \frac{3\, z^2 W(x_0)^{-n} \overline{z}^2}{16\,\epsilon k^2 \cosh(x_0)^3 \sinh(x_0)^3} + \frac{n^2 W(x_0)^{-n}}{4\,\epsilon k^2 \cosh(x_0) \sinh(x_0)^3} \\
& + \frac{z^2 W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)}{8\,\epsilon k^2 W(x_0) \cosh(x_0)^2 \sinh(x_0)^2} + \frac{W(x_0)^{-n} \overline{z}^2 \frac{\partial}{\partial x_0} W(x_0)}{8\,\epsilon k^2 W(x_0) \cosh(x_0)^2 \sinh(x_0)^2} + \frac{z W(x_0)^{-n} \overline{z} \frac{\partial}{\partial x_0} W(x_0)^2}{8\,\epsilon k^2 W(x_0)^2 \cosh(x_0) \sinh(x_0)^3} - \frac{W(x_0)^{-n} \cosh(x_0) \frac{\partial}{\partial x_0} W(x_0)^2}{16\,\epsilon k^2 W(x_0)^2 \sinh(x_0)^3} \\
& - \frac{n W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)}{8\,\epsilon k^2 W(x_0) \cosh(x_0)^2 \sinh(x_0)^2} + \frac{3\, z W(x_0)^{-n} \overline{z}}{8\,\epsilon k^2 \cosh(x_0)^3 \sinh(x_0)^3} + \frac{W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)^2}{16\,\epsilon k^2 W(x_0)^2 \cosh(x_0) \sinh(x_0)^3} + \frac{3\, W(x_0)^{-n}}{16\,\epsilon k^2 \cosh(x_0)^3 \sinh(x_0)^3}
\end{aligned}
\tag{141}
$$

$$
\begin{aligned}
\Phi'_{01} = & -\frac{n^2 z^2 W(x_0)^{-n} \overline{z} \cosh(x_0)}{4\,\epsilon k^2 \sinh(x_0)^3} + \frac{n z^2 W(x_0)^{-n} \overline{z} \frac{\partial}{\partial x_0} W(x_0)}{4\,\epsilon k^2 W(x_0) \sinh(x_0)^2} + \frac{n^2 z^2 W(x_0)^{-n} \overline{z}}{4\,\epsilon k^2 \cosh(x_0) \sinh(x_0)^3} + \frac{n^2 z W(x_0)^{-n} \cosh(x_0)}{4\,\epsilon k^2 \sinh(x_0)^3} - \frac{z^2 W(x_0)^{-n} \overline{z} \cosh(x_0) \frac{\partial}{\partial x_0} W(x_0)^2}{16\,\epsilon k^2 W(x_0)^2 \sinh(x_0)^3} \\
& - \frac{n W(x_0)^{-n} \overline{z}}{2\,\epsilon k^2 \cosh(x_0) \sinh(x_0)} - \frac{n z^2 W(x_0)^{-n} \overline{z} \frac{\partial}{\partial x_0} W(x_0)}{8\,\epsilon k^2 W(x_0) \cosh(x_0)^2 \sinh(x_0)^2} + \frac{n z W(x_0)^{-n} \cosh(x_0) \frac{\partial}{\partial x_0} W(x_0)^2}{8\,\epsilon k^2 W(x_0)^2 \sinh(x_0)^3} - \frac{n z W(x_0)^{-n} \cosh(x_0) \frac{\partial^2}{(\partial x_0)^2} W(x_0)}{8\,\epsilon k^2 W(x_0) \sinh(x_0)^3} \\
& - \frac{n^2 z W(x_0)^{-n}}{4\,\epsilon k^2 \cosh(x_0) \sinh(x_0)^3} + \frac{W(x_0)^{-n} \overline{z} \frac{\partial}{\partial x_0} W(x_0)}{4\,\epsilon k^2 W(x_0) \cosh(x_0)^2} + \frac{z^2 W(x_0)^{-n} \overline{z} \frac{\partial}{\partial x_0} W(x_0)^2}{16\,\epsilon k^2 W(x_0)^2 \cosh(x_0) \sinh(x_0)^3} - \frac{z W(x_0)^{-n} \cosh(x_0) \frac{\partial}{\partial x_0} W(x_0)^2}{16\,\epsilon k^2 W(x_0)^2 \sinh(x_0)^3} \\
& - \frac{W(x_0)^{-n} \overline{z} \frac{\partial}{\partial x_0} W(x_0)^2}{8\,\epsilon k^2 W(x_0)^2 \cosh(x_0) \sinh(x_0)} + \frac{W(x_0)^{-n} \overline{z} \frac{\partial^2}{(\partial x_0)^2} W(x_0)}{8\,\epsilon k^2 W(x_0) \cosh(x_0) \sinh(x_0)} - \frac{n z W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)^2}{8\,\epsilon k^2 W(x_0)^2 \cosh(x_0) \sinh(x_0)^3} + \frac{n z W(x_0)^{-n} \frac{\partial^2}{(\partial x_0)^2} W(x_0)}{8\,\epsilon k^2 W(x_0) \cosh(x_0) \sinh(x_0)^3} \\
& + \frac{3\, z^2 W(x_0)^{-n} \overline{z}}{16\,\epsilon k^2 \cosh(x_0)^3 \sinh(x_0)^3} + \frac{W(x_0)^{-n} \overline{z} \frac{\partial}{\partial x_0} W(x_0)}{8\,\epsilon k^2 W(x_0) \cosh(x_0)^2 \sinh(x_0)^2} + \frac{z W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)^2}{16\,\epsilon k^2 W(x_0)^2 \cosh(x_0) \sinh(x_0)^3} + \frac{3\, z W(x_0)^{-n}}{16\,\epsilon k^2 \cosh(x_0)^3 \sinh(x_0)^3}
\end{aligned}
\tag{142}
$$

$$
\begin{aligned}
\Phi'_{02} = & -\frac{n^2 z^2 W(x_0)^{-n} \cosh(x_0)}{4\,\epsilon k^2 \sinh(x_0)^3} + \frac{n z^2 W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)}{4\,\epsilon k^2 W(x_0) \sinh(x_0)^2} + \frac{n^2 z^2 W(x_0)^{-n}}{4\,\epsilon k^2 \cosh(x_0) \sinh(x_0)^3} - \frac{z^2 W(x_0)^{-n} \cosh(x_0) \frac{\partial}{\partial x_0} W(x_0)^2}{16\,\epsilon k^2 W(x_0)^2 \sinh(x_0)^3} \\
& - \frac{n W(x_0)^{-n}}{2\,\epsilon k^2 \cosh(x_0) \sinh(x_0)} - \frac{n z^2 W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)}{8\,\epsilon k^2 W(x_0) \cosh(x_0)^2 \sinh(x_0)^2} + \frac{W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)}{4\,\epsilon k^2 W(x_0) \cosh(x_0)^2} + \frac{z^2 W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)^2}{16\,\epsilon k^2 W(x_0)^2 \cosh(x_0) \sinh(x_0)^3} \\
& - \frac{W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)^2}{8\,\epsilon k^2 W(x_0)^2 \cosh(x_0) \sinh(x_0)} + \frac{W(x_0)^{-n} \frac{\partial^2}{(\partial x_0)^2} W(x_0)}{8\,\epsilon k^2 W(x_0) \cosh(x_0) \sinh(x_0)} + \frac{3\, z^2 W(x_0)^{-n}}{16\,\epsilon k^2 \cosh(x_0)^3 \sinh(x_0)^3} + \frac{W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)}{8\,\epsilon k^2 W(x_0) \cosh(x_0)^2 \sinh(x_0)^2}
\end{aligned}
\tag{143}
$$

$$
\begin{aligned}
\Phi'_{10} = & -\frac{n^2 z W(x_0)^{-n} \overline{z}^2 \cosh(x_0)}{4\,\epsilon k^2 \sinh(x_0)^3} + \frac{n z W(x_0)^{-n} \overline{z}^2 \frac{\partial}{\partial x_0} W(x_0)}{4\,\epsilon k^2 W(x_0) \sinh(x_0)^2} + \frac{n^2 z W(x_0)^{-n} \overline{z}^2}{4\,\epsilon k^2 \cosh(x_0) \sinh(x_0)^3} + \frac{n^2 W(x_0)^{-n} \overline{z} \cosh(x_0)}{4\,\epsilon k^2 \sinh(x_0)^3} - \frac{z W(x_0)^{-n} \overline{z}^2 \cosh(x_0) \frac{\partial}{\partial x_0} W(x_0)^2}{16\,\epsilon k^2 W(x_0)^2 \sinh(x_0)^3} \\
& - \frac{n z W(x_0)^{-n}}{2\,\epsilon k^2 \cosh(x_0) \sinh(x_0)} - \frac{n z W(x_0)^{-n} \overline{z}^2 \frac{\partial}{\partial x_0} W(x_0)}{8\,\epsilon k^2 W(x_0) \cosh(x_0)^2 \sinh(x_0)^2} + \frac{n W(x_0)^{-n} \overline{z} \cosh(x_0) \frac{\partial}{\partial x_0} W(x_0)^2}{8\,\epsilon k^2 W(x_0)^2 \sinh(x_0)^3} - \frac{n W(x_0)^{-n} \overline{z} \cosh(x_0) \frac{\partial^2}{(\partial x_0)^2} W(x_0)}{8\,\epsilon k^2 W(x_0) \sinh(x_0)^3} \\
& - \frac{n^2 W(x_0)^{-n} \overline{z}}{4\,\epsilon k^2 \cosh(x_0) \sinh(x_0)^3} + \frac{z W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)}{4\,\epsilon k^2 W(x_0) \cosh(x_0)^2} + \frac{z W(x_0)^{-n} \overline{z}^2 \frac{\partial}{\partial x_0} W(x_0)^2}{16\,\epsilon k^2 W(x_0)^2 \cosh(x_0) \sinh(x_0)^3} - \frac{W(x_0)^{-n} \overline{z} \cosh(x_0) \frac{\partial}{\partial x_0} W(x_0)^2}{16\,\epsilon k^2 W(x_0)^2 \sinh(x_0)^3} \\
& - \frac{z W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)^2}{8\,\epsilon k^2 W(x_0)^2 \cosh(x_0) \sinh(x_0)} + \frac{z W(x_0)^{-n} \frac{\partial^2}{(\partial x_0)^2} W(x_0)}{8\,\epsilon k^2 W(x_0) \cosh(x_0) \sinh(x_0)} - \frac{n W(x_0)^{-n} \overline{z} \frac{\partial}{\partial x_0} W(x_0)^2}{8\,\epsilon k^2 W(x_0)^2 \cosh(x_0) \sinh(x_0)^3} + \frac{n W(x_0)^{-n} \overline{z} \frac{\partial^2}{(\partial x_0)^2} W(x_0)}{8\,\epsilon k^2 W(x_0) \cosh(x_0) \sinh(x_0)^3} \\
& + \frac{3\, z W(x_0)^{-n} \overline{z}^2}{16\,\epsilon k^2 \cosh(x_0)^3 \sinh(x_0)^3} + \frac{z W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)}{8\,\epsilon k^2 W(x_0) \cosh(x_0)^2 \sinh(x_0)^2} + \frac{W(x_0)^{-n} \overline{z} \frac{\partial}{\partial x_0} W(x_0)^2}{16\,\epsilon k^2 W(x_0)^2 \cosh(x_0) \sinh(x_0)^3} + \frac{3\, W(x_0)^{-n} \overline{z}}{16\,\epsilon k^2 \cosh(x_0)^3 \sinh(x_0)^3}
\end{aligned}
\tag{144}
$$

$$
\begin{aligned}
\Phi'_{11} = & -\frac{n^2 z W(x_0)^{-n} \overline{z} \cosh(x_0)}{4\,\epsilon k^2 \sinh(x_0)^3} + \frac{n z W(x_0)^{-n} \overline{z} \frac{\partial}{\partial x_0} W(x_0)}{4\,\epsilon k^2 W(x_0) \sinh(x_0)^2} + \frac{n^2 z W(x_0)^{-n} \overline{z}}{4\,\epsilon k^2 \cosh(x_0) \sinh(x_0)^3} + \frac{n^2 W(x_0)^{-n} \cosh(x_0)}{8\,\epsilon k^2 \sinh(x_0)^3} - \frac{z W(x_0)^{-n} \overline{z} \cosh(x_0) \frac{\partial}{\partial x_0} W(x_0)^2}{16\,\epsilon k^2 W(x_0)^2 \sinh(x_0)^3} \\
& - \frac{n z W(x_0)^{-n} \overline{z} \frac{\partial}{\partial x_0} W(x_0)}{8\,\epsilon k^2 W(x_0) \cosh(x_0)^2 \sinh(x_0)^2} + \frac{n W(x_0)^{-n} \cosh(x_0) \frac{\partial}{\partial x_0} W(x_0)^2}{16\,\epsilon k^2 W(x_0)^2 \sinh(x_0)^3} - \frac{n W(x_0)^{-n} \cosh(x_0) \frac{\partial^2}{(\partial x_0)^2} W(x_0)}{16\,\epsilon k^2 W(x_0) \sinh(x_0)^3} \\
& - \frac{n^2 W(x_0)^{-n}}{8\,\epsilon k^2 \cosh(x_0) \sinh(x_0)^3} + \frac{z W(x_0)^{-n} \overline{z} \frac{\partial}{\partial x_0} W(x_0)^2}{16\,\epsilon k^2 W(x_0)^2 \cosh(x_0) \sinh(x_0)^3} - \frac{W(x_0)^{-n} \cosh(x_0) \frac{\partial}{\partial x_0} W(x_0)^2}{32\,\epsilon k^2 W(x_0)^2 \sinh(x_0)^3} - \frac{n W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)^2}{16\,\epsilon k^2 W(x_0)^2 \cosh(x_0) \sinh(x_0)^3} \\
& + \frac{n W(x_0)^{-n} \frac{\partial^2}{(\partial x_0)^2} W(x_0)}{16\,\epsilon k^2 W(x_0) \cosh(x_0) \sinh(x_0)^3} + \frac{3\, z W(x_0)^{-n} \overline{z}}{16\,\epsilon k^2 \cosh(x_0)^3 \sinh(x_0)^3} + \frac{W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)^2}{32\,\epsilon k^2 W(x_0)^2 \cosh(x_0) \sinh(x_0)^3} + \frac{3\, W(x_0)^{-n}}{32\,\epsilon k^2 \cosh(x_0)^3 \sinh(x_0)^3}
\end{aligned}
\tag{145}
$$

$$\Phi'_{12} = -\frac{n^2 z W(x_0)^{-n}\cosh(x_0)}{4\,\epsilon k^2 \sinh(x_0)^3} + \frac{n z W(x_0)^{-n}\frac{\partial}{\partial x_0}W(x_0)}{4\,\epsilon k^2 W(x_0)\sinh(x_0)^2} + \frac{n^2 z W(x_0)^{-n}}{4\,\epsilon k^2\cosh(x_0)\sinh(x_0)^3} - \frac{z W(x_0)^{-n}\cosh(x_0)\frac{\partial}{\partial x_0}W(x_0)^2}{16\,\epsilon k^2 W(x_0)^2\sinh(x_0)^3}$$
$$- \frac{n z W(x_0)^{-n}\frac{\partial}{\partial x_0}W(x_0)}{8\,\epsilon k^2 W(x_0)\cosh(x_0)^2\sinh(x_0)^2} + \frac{z W(x_0)^{-n}\frac{\partial}{\partial x_0}W(x_0)^2}{16\,\epsilon k^2 W(x_0)^2\cosh(x_0)\sinh(x_0)^3} + \frac{3\,z W(x_0)^{-n}}{16\,\epsilon k^2\cosh(x_0)^3\sinh(x_0)^3} \tag{146}$$

$$\Phi'_{20} = -\frac{n^2 W(x_0)^{-n}\overline{z}^2\cosh(x_0)}{4\,\epsilon k^2\sinh(x_0)^3} + \frac{n W(x_0)^{-n}\overline{z}^2\frac{\partial}{\partial x_0}W(x_0)}{4\,\epsilon k^2 W(x_0)\sinh(x_0)^2} + \frac{n^2 W(x_0)^{-n}\overline{z}^2}{4\,\epsilon k^2\cosh(x_0)\sinh(x_0)^3} - \frac{W(x_0)^{-n}\overline{z}^2\cosh(x_0)\frac{\partial}{\partial x_0}W(x_0)^2}{16\,\epsilon k^2 W(x_0)^2\sinh(x_0)^3}$$
$$- \frac{n W(x_0)^{-n}}{2\,\epsilon k^2\cosh(x_0)\sinh(x_0)} - \frac{n W(x_0)^{-n}\overline{z}^2\frac{\partial}{\partial x_0}W(x_0)}{8\,\epsilon k^2 W(x_0)\cosh(x_0)^2\sinh(x_0)^2} + \frac{W(x_0)^{-n}\frac{\partial}{\partial x_0}W(x_0)}{4\,\epsilon k^2 W(x_0)\cosh(x_0)^2} + \frac{W(x_0)^{-n}\overline{z}^2\frac{\partial}{\partial x_0}W(x_0)^2}{16\,\epsilon k^2 W(x_0)^2\cosh(x_0)\sinh(x_0)^3} \tag{147}$$
$$- \frac{W(x_0)^{-n}\frac{\partial}{\partial x_0}W(x_0)^2}{8\,\epsilon k^2 W(x_0)^2\cosh(x_0)\sinh(x_0)} + \frac{W(x_0)^{-n}\frac{\partial^2}{(\partial x_0)^2}W(x_0)}{8\,\epsilon k^2 W(x_0)\cosh(x_0)\sinh(x_0)} + \frac{3\,W(x_0)^{-n}\overline{z}^2}{16\,\epsilon k^2\cosh(x_0)^3\sinh(x_0)^3} + \frac{W(x_0)^{-n}\frac{\partial}{\partial x_0}W(x_0)}{8\,\epsilon k^2 W(x_0)\cosh(x_0)^2\sinh(x_0)^2}$$

$$\Phi'_{21} = -\frac{n^2 W(x_0)^{-n}\overline{z}\cosh(x_0)}{4\,\epsilon k^2\sinh(x_0)^3} + \frac{n W(x_0)^{-n}\overline{z}\frac{\partial}{\partial x_0}W(x_0)}{4\,\epsilon k^2 W(x_0)\sinh(x_0)^2} + \frac{n^2 W(x_0)^{-n}\overline{z}}{4\,\epsilon k^2\cosh(x_0)\sinh(x_0)^3} - \frac{W(x_0)^{-n}\overline{z}\cosh(x_0)\frac{\partial}{\partial x_0}W(x_0)^2}{16\,\epsilon k^2 W(x_0)^2\sinh(x_0)^3}$$
$$- \frac{n W(x_0)^{-n}\overline{z}\frac{\partial}{\partial x_0}W(x_0)}{8\,\epsilon k^2 W(x_0)\cosh(x_0)^2\sinh(x_0)^2} + \frac{W(x_0)^{-n}\overline{z}\frac{\partial}{\partial x_0}W(x_0)^2}{16\,\epsilon k^2 W(x_0)^2\cosh(x_0)\sinh(x_0)^3} + \frac{3\,W(x_0)^{-n}\overline{z}}{16\,\epsilon k^2\cosh(x_0)^3\sinh(x_0)^3} \tag{148}$$

$$\Phi'_{22} = \Phi_{22} \tag{149}$$

$$\Lambda' = \Lambda \tag{150}$$

# References


[1] H. Stephani, D. Kramer, M. MacCallum, C. Hoenselaers, and E. Herlt, Exact Solutions of Einstein's Field Equations. Cambridge Monographs on Mathematical Physics, Cambridge University Press, 2009.

**Calculated by:**

*Name: Mustafa Tosun*

*E-mail:*

*Date:*